Precambrian Research 196–197 (2012) 1–22

Contents lists available at SciVerse ScienceDirect

Precambrian Research

journal homepage: www.elsevier.com/locate/precamres

Tectonic implications of the 1419 Ma Nova Guarita mafic intrusives paleomagnetic pole (Amazonian Craton) on the longevity of Nuna

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ARTICLE INFO

Article history:
Received 1 April 2011
Received in revised form 24 October 2011
Accepted 28 October 2011
Available online xxx

Keywords:
Paleomagnetism
^{40}Ar–^{39}Ar geochronology
Nova Guarita dykes
Amazonian Craton
Nuna Supercontinent

ABSTRACT

The Nuna supercontinent was probably assembled in the Paleoproterozoic, but its paleogeography and the timing for its demise are still a matter of debate. A paleomagnetic and geochronological study carried out on the Mesoproterozoic Nova Guarita dyke swarm (northern Mato Grosso State, SW Amazonian Craton) provides additional constraints on the duration of this supercontinent. Paleomagnetic AF and thermal treatment revealed south/southwest (northeast) magnetic directions with downward (upward) inclinations for 19 analyzed sites. These directions are carried by PSD magnetite with high unblocking temperatures as indicated by additional magnetic tests, including thermomagnetic curves, hysteresis loops and the progressive acquisition of isothermal remanence in selected samples. A positive contact test with the host granite in one of the studied dykes further attests to the primary origin of the characteristic magnetic component. A mean site direction was calculated at $D_m = 220.5^\circ$, $I_m = 45.9^\circ$ ($\alpha_{95} = 6.5^\circ$, $K = 27.7$), which yielded a paleomagnetic pole located at 245.9°E, 47.9°S ($A_{95} = 7.0^\circ$). $^{40}Ar/^{39}Ar$ dating carried out on biotites from four analyzed dykes yielded well-defined plateau ages with a mean of 1418.5 ± 3.5 Ma. The Nova Guarita pole precludes a long-lived Nuna configuration in which Laurentia, Baltica, North China, and Amazonia formed a long and continuous block as previously proposed for the Paleoproterozoic. It is nevertheless fully compatible with a SAMBA (Amazonia-Baltica) link at Mesoproterozoic times.

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1. Introduction

A Paleoproterozoic supercontinent has been suggested by several authors (Rogers, 1996; Rogers and Santosh, 2002; Meert, 2002; Zhao et al., 2002, 2004; Hou et al., 2008a,b). This supercontinent has been referred in the literature as either Columbia or Nuna (Hoffman, 1997; Rogers and Santosh, 2002). Here we favor 'Nuna' following the reasoning of Reddy and Evans (2009) for the precedence of this name over Columbia. In spite of the great amount of work on the paleogeography of Nuna, two major issues still remain: (i) which continental blocks took part in Nuna and how were they arranged? (ii) Did Nuna stand as a great continental mass for a long time or was it ephemeral? One way to answer these questions is through systematic paleomagnetic studies of Paleoproterozoic geological units worldwide, preferentially on the largest continental cratonic blocks which may have been part of Nuna, such as Laurentia, Baltica, North China, Siberia and Amazonia. Unfortunately, the Paleo-Mesoproterozoic paleomagnetic dataset is still very scant, which hamper a complete paleomagnetic test of Nuna to be performed (Pesonen et al., 2003; Meert and Torsvik, 2003).

Yet, some progress has been done in recent years with publication of new high quality paleomagnetic data from some of these continental blocks. For example, Salminen and Pesonen (2007) and Lubnina et al. (2010) demonstrate that Laurentia and Baltica drifted together during most of Mesoproterozoic (1760–1260 Ma), and that this continental block probably formed the core of the Nuna supercontinent. This continental landmass is expected to have agglutinated at ca. 1800 Ma when the Ukrainian Shield accreted to Fenoscandia forming Baltica (Elming et al., 2010), and the Hae–Hearne–Slave craton accreted to Superior craton along the Trans-Hudson suture forming Laurentia (Hoffman, 1988; Irving et al., 2004). In most Nuna reconstructions, the Amazonian Craton is considered to be linked to Baltica based on geological evidence (e.g., Geraldes et al., 2001; Rogers and Santosh, 2002; Zhao et al.,

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doi:10.1016/j.precamres.2011.10.022

2002, 2004; Johansson, 2009). However, in a recent work Bispo-Santos et al. (2008) have used key paleomagnetic poles at ~1780 Ma to propose a paleogeographic reconstruction in which Amazonia is linked to North China and form the southern tip of a long and continuous landmass comprising also Laurentia and Baltica linked by Paleo- to Mesoproterozoic mobile belts (see also Cordani et al., 2009). In their model, Amazonia and North China were linked through their Ventuari-Tapajós (e.g., Tassinari et al., 2000; Cordani and Teixeira, 2007) and Trans-North China (e.g., Zhao et al., 2005, 2006) mobile belts, respectively. While juvenile magmatic arcs were being accreted to the southwestern border of the Amazonian Craton between 1950 and 1800 Ma, forming the Ventuari-Tapajós belt, in the North China Craton, the Western Block was colliding with the Eastern Block forming the Trans-North China belt, which culminated with the formation of the North China Craton at about 1850 Ma. A link between Baltica, North China and Amazonia was already proposed by Kusky et al. (2007), based on geological grounds. These authors suggested also that North China disrupted from Nuna soon after its formation at about 1850 Ma, indicated by expressive mafic dike swarms that intruded the northern and southern North China at ~1780 Ma. If so, the Amazonian Craton also disrupted from Nuna at the same time. Dykes and sills of similar age are also described in Brazil (Santos et al., 2003) and in Venezuela (Onstott et al., 1984).

In this paper we address the question of Nuna's lifetime in the configuration proposed by Bispo-Santos et al. (2008) – was it long or short-lived? – by studying a recently discovered dyke swarm in central Amazonian craton. This new dyke swarm, baptized here as Nova Guarita Intrusive Suite from the name of a nearby village, was sampled for paleomagnetic and geocrhonological purposes and the results presented hereafter suggest that the Amazonian Craton was not part of Nuna by 1420 Ma ago. Another Mesoproterozoic reconstruction – the SAMBA configuration of Johansson (2009) – is tested using our new pole and the available paleomagnetic data for Baltica and Laurentia.

2. Geological setting

Considered as one of the largest cratonic areas in the world with about 430,000 km^2, the Amazonian Craton is bisected by the E–W trending Paleozoic Amazonian sedimentary basin (Fig. 1). It is composed of two small Archaean cores (the Central Amazonian Province, with ages >2600 Ma) encircled by predominantly accretionary Paleoproterozoic belts of the 2250–2050 Ma Maroni-Itacaiunas Province, the 1980–1810 Ma Ventuari-Tapajós Province, the 1780–1550 Ma Rio Negro-Juruena Province, the 1550–1300 Ma Rondonian-San Ignacio Province, and the ca. 1250–950 Ma Sunsás Province (Tassinari et al., 2000; Cordani and Teixeira, 2007; Cordani et al., 2009).

The study area is located in northern Mato Grosso State, and geotectonically inserted into the Ventuari-Tapajos Province (Fig. 1). This province is predominantly composed by granitic-gneissic rocks metamorphosed at amphibolite facies. These rocks were dated at 1980 and 1810 Ma by Rb–Sr and U–Pb methods, which are coincident with or slightly younger than the respective Sm–Nd model ages (Cordani and Teixeira, 2007). Together with the positive $\xi_{ND(T)}$ values, this suggests a juvenile nature for these rocks. In its southwestern boundary the province is bordered by the younger 1780–1550 Ma Rio Negro-Juruena juvenile mobile belt, composed by gneisses, granodiorites, tonalites, migmatites, granites and amphibolites (Fig. 1).

Here we assign the name Nova Guarita Intrusive Suite to the profuse mafic dykes that crop out around the Nova Guarita village, mainly along the road (MT-208) between Nova Guarita and Terra Nova city (northern Mato Grosso State). These basic dykes are represented by gabbros, diabases and microgabbros cutting the granites from the Matupa intrusive Suite, Teles Pires and São Pedro Granites, as well as the felsic rocks from the Colider volcanic intrusive Suite. All these units present ages between 1700 and 1900 Ma (Moreton and Martins, 2005). The dykes strike predominantly NW (some with NE trend), and range from one meter up to several meters in thickness. The first records of these dykes are attributed to Silva et al. (1980), who obtained a whole rock K–Ar age of 1416 ± 14 Ma for a diabase (not shown in Fig. 1) located in the extreme northwest of the 'Folha Ilha 24 de Maio' (Frasca and Borges, 2005). Interestingly, another dyke swarm within a similar geological context but previously interpreted as Paleoproterozoic in age occur around Peixoto de Azevedo and Matupa towns (Fig. 1), cut the Matupa Intrusive Suite (1848 ± 17, U–Pb) and the Flor da Serra mafic to intermediate igneous suite (Lacerda-Filho et al., 2004).

3. Sampling and methods

Standard oriented cylindrical cores were collected from 19 sites comprising diabases, gabbros and microgabbros from the Nova Guarita intrusive suite (Fig. 1). Samples were oriented using both sun and magnetic compasses. To investigate the magnetic stability of ChRM directions, and their origin, a baked-contact test was performed for site 5 (SD74 samples), where the mafic dyke cuts granitic rocks from the Matupa intrusive Suite. From the exposition of this particular dyke a thickness of more than 20 m is suggested, although the contact with the granite is not visible. For this paleomagnetic test, besides the 10 cylindrical cores from the intrusion (samples SD74A-J), the granite was sampled at 1.90 m (sample SD74K), at 2.17 m (sample SD74L), and at approximately 55 m from the contact, where 5 cylindrical cores were collected from a large granitic outcrop (SD91A-E samples).

Paleomagnetic analysis was carried out in the paleomagnetic laboratories at the University of São Paulo (USP), Brazil and the Luleå University of Technology, Sweden. Step-wise alternating magnetic field (AF) and thermal demagnetization techniques were employed to separate the characteristic remanent magnetization (ChRM) component carried by the samples. Normally, steps of 2.5 mT (up to 15 mT) and 5 mT (15–100 mT) were employed for AF demagnetization using a 2-axis tumbler Molspin AF demagnetizer, or an automated three-axis AF-demagnetizer in conjunction with a cryogenic magnetometer. Thermal demagnetization was performed using a Magnetic Measurements TD-60 furnace in steps of 50 °C (from 150 °C up to 500 °C) and 20 °C (from 500 °C up to 600 °C). Remanent magnetization measurements were carried out using a portable slow-speed fluxgate Molspin magnetometer (MINISPIN) or a 2-G cryogenic magnetometer. In São Paulo these instruments are located in a magnetically shielded room with ambient field <1000 nT. Magnetic components for each specimen were identified in orthogonal plots (Zijderveld, 1967), and calculated using the principal component analysis (Kirshvink, 1980). At least 4 demagnetization steps were used to calculate vectors using least-squares fits, and an upper limit for mean angular deviation (MAD) of 8° was used. Fisher's (1953) statistics was used to calculate site mean directions and the paleomagnetic pole.

Magnetic mineralogy was investigated through acquisition of isothermal remanent magnetization (IRM) in a Magnetic Measurements pulse magnetizer, and hysteresis curves using a MicroMag Vibrating Sample Magnetometer (Model 3900-Princeton Measurements Corporation) from the University of Southampton – National Oceonography Centre (NOC, England). These curves give bulk coercive force (H_c), saturation magnetization (M_s), and saturation isothermal remanence (M_{rs}), after subtraction of the paramagnetic contribution from the high field portion of the curve. Coercivity of remanence (H_{cr}) was obtained using the same VSM instrument.

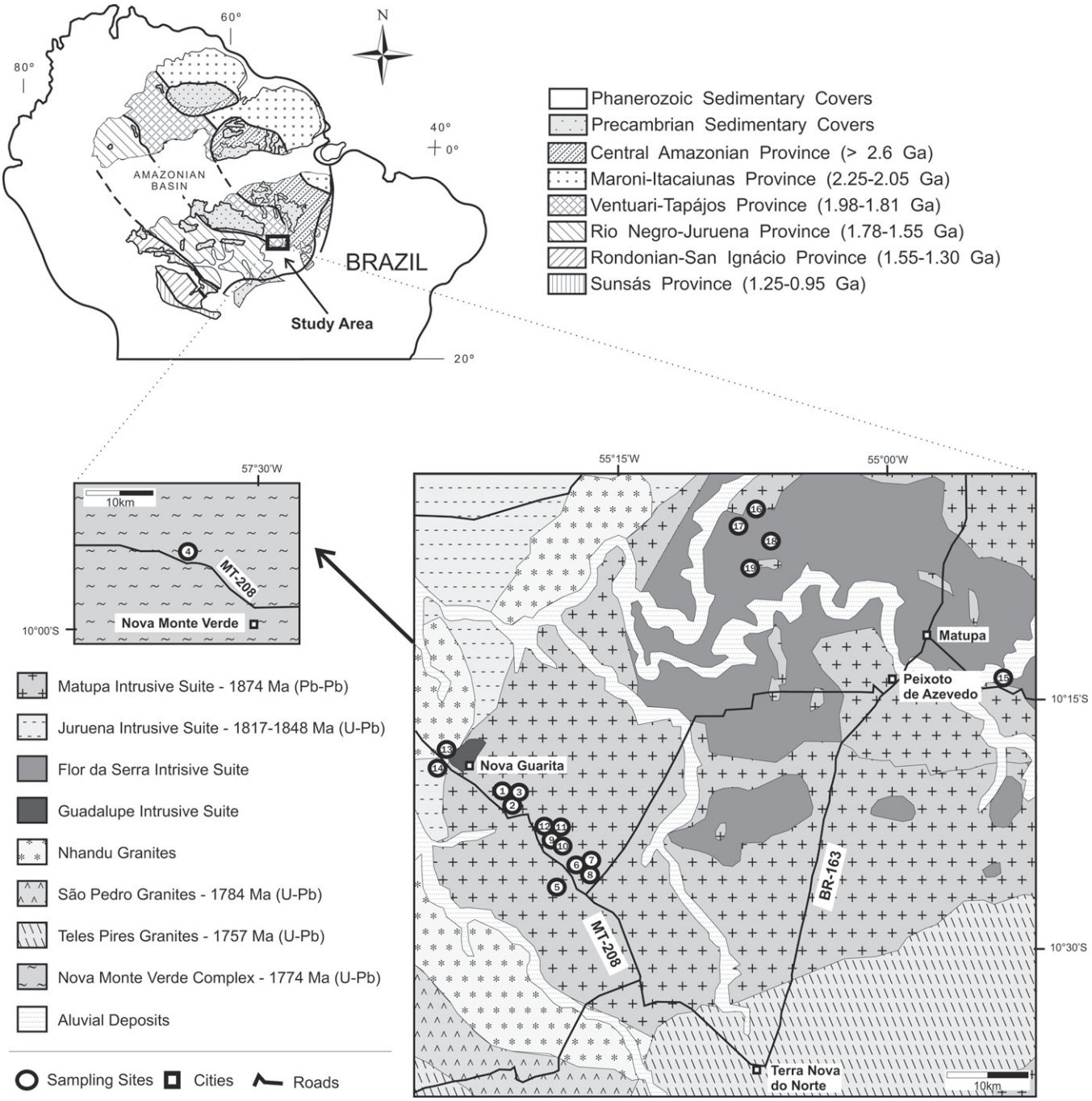


Fig. 1. Geological map of the studied area with location of sampling sites (circled numbers) (after Lacerda-Filho et al., 2004). Inset – Amazonian Craton and their geochronological Provinces (after Tassinari et al., 2000).

Also, thermomagnetic curves (low-field magnetic susceptibility versus (high and low) temperature) were performed for several samples to characterize the magnetic carriers in the rock, using a CS-4 apparatus coupled to the KLY-4S Kappabridge instrument (Agico, Czech Republic). Using a Kappabridge the anisotropy of magnetic susceptibility (AMS) was determined to define the magnetic fabric in the rocks. The technique of Jelinek (1981) was used for the statistical analyses of the AMS data.

4. Petrography

Thin and polished sections of several dykes were examined petrographically using transmitted and reflected light microscopy. The basic rocks of the Nova Guarita dykes present ophitic to subophitic textures, and consist of plagioclase crystals (30–35%) partly saussuritized (up to 15%), augite (15–25%), titanoaugite (20–10%), hypersthene (5–15%), euhedral biotite (5–10%), and epidote (<5–10%) appearing as alteration rims in several minerals (Plate 1A). Biotite occurs usually associated to opaque phases (Plate 1B).

In polished sections, medium to coarse grains (up to 500 μm) of intergrown magnetite and ilmenite are observed, the latter occurring as parallel pod-like lamellae in the host magnetite (Plate 1C). Occasionally, ilmenite appears as coarser subgrain domains (Plate 1D). These features are inferred to result from high-temperature oxidation/exsolution upon initial cooling of the rock. In a few cases,

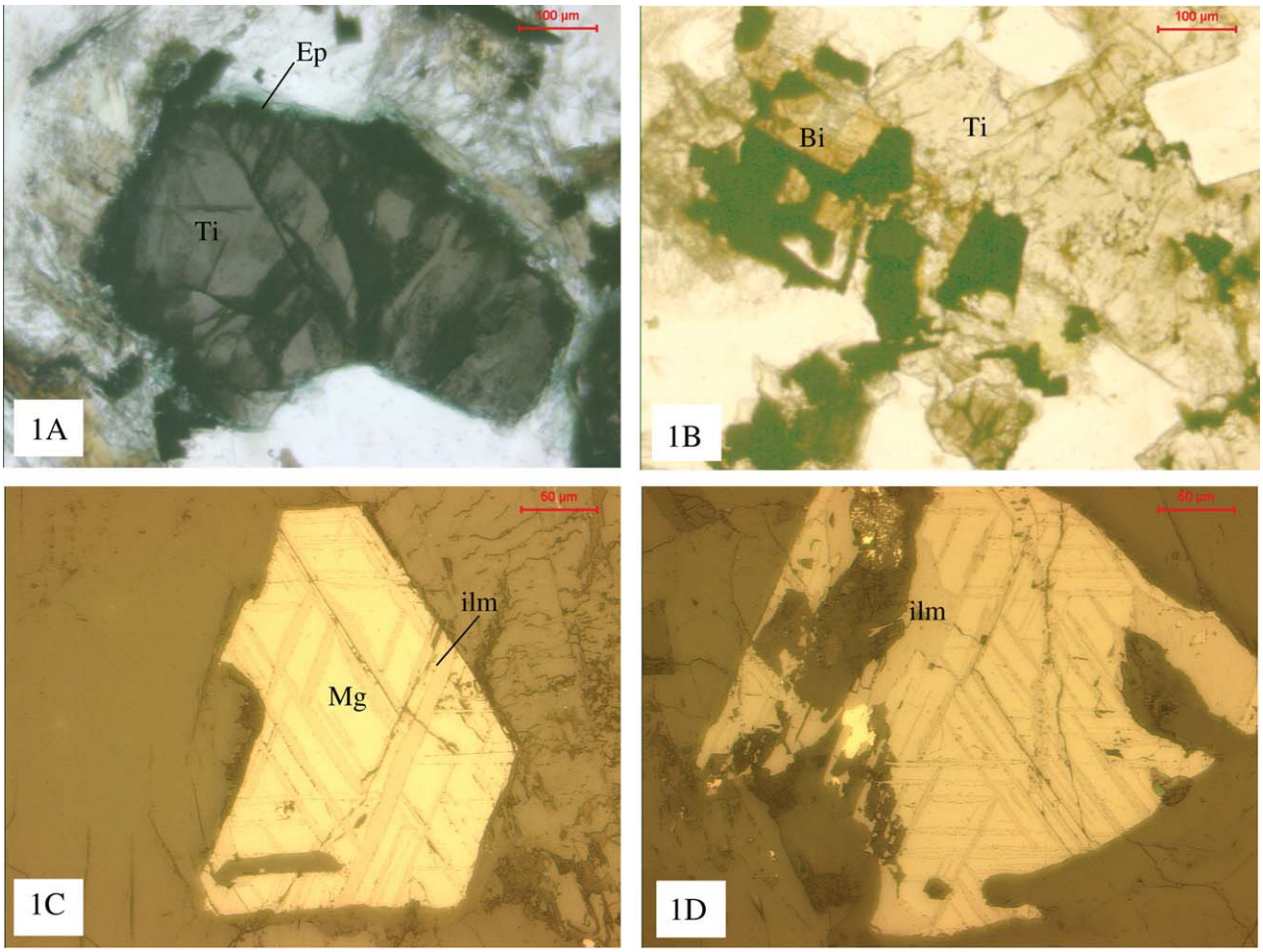


Plate 1. Photomicrograph of basic rocks from the Nova Guarita dykes unit. (A) (sample SD74-H2): Epidote (E_p) rim arround titanoaugite (Ti) crystal; (B) (sample SD74-G1): Biotite (Bi) replacing titanoaugite and opaque borders; (C) (sample SD78-G1): Magnetite (Mg) with thick lamellae of ilmenite (ilm); (D) (sample SD83-B1): Magnetite associated with ilmenite. Plates A and B are transmitted light microscopy. Plates C and D are reflected light microscopy. All photos are with parallel polarizers.

magnetite is found partially martitized. Unaltered crystals of magnetite of smaller sizes (up to 300 μm) are also observed (Plate 2A and B). Small hematite crystals (Plate 2C) are sometimes locally present, as magnetite pseudomorphs (Plate 2D). In one sample, intergranular growth of hematite was observed (Plate 2E). Small crystals of hematite are associated with magnetite with ilmenite lamellae (Plate 2F).

5. Paleomagnetism

The intensity of natural remanent magnetization (NRM) ranges from 0.2 A/m up to values as high as 761 A/m. These intensities are typical of basaltic rocks with high contribution of ferromagnetic minerals. Thermal demagnetization performed on pilot samples in the steps described above (Section 3) disclosed the characteristic remanent magnetization (ChRM) for only 33% of samples (Fig. 2). This treatment shows that this component is carried by magnetic grains with high-unblocking temperatures (T_{ub}) between 500–520 °C and 580 °C. The AF demagnetization was much more efficient to isolate the ChRM directions, yielding meaningful remanent directions for specimens from all 19 analyzed sites. Normally, scattered low-coercivity secondary components are removed in fields up to 30–50 mT leaving high-coercivity (H_c) south/southwest (northeast) directions with downward (upward) inclinations (Fig. 3), which was named *N*ova *G*uarita component (*NG* component).

Table 1 presents a summary of paleomagnetic results. Site mean directions are shown in Fig. 4, which cluster around the mean, $D_m = 220.5°$, $I_m = 45.9°$ ($\alpha_{95} = 6.5°$, $K = 27.7$, $N = 19$). The mean of VGPs yielded the paleomagnetic NG (Nova Guarita) pole located at 245.9°E, 47.9°S ($A_{95} = 7.0°$, $K = 23.7$).

A baked-contact test was performed on specimens from site 5 (SD74 samples), where a dyke from the Nova Guarita suite cuts granitic rocks from the Matupa intrusive Suite (see Section 3). After AF demagnetization the two host rock samples (SD74K and SD74L) collected within a distance <2.5 m from the dyke, show southwestern directions with downward inclinations (Fig. 5b and c), similar to the high-H_c components isolated in the dyke itself (Fig. 5a), and designated as NG component. Samples at ~55 m from the dyke contact show northeastern directions with downward inclinations (Fig. 5d). Fig. 6 shows sample mean directions for the baked contact test performed for site 5. In the samples from the granitic outcrop situated 55 m away from the contact the magnetic components are very different from those obtained for the dyke itself, suggesting that no posterior regional remagnetization occurred in the dykes, and that the high H_c and T_{ub} components carried by them represents a primary thermo-remanent magnetization acquired during cooling of the rock.

6. Magnetic mineralogy

AF demagnetization shows differences in coercivity between specimens, although their decay in magnetization intensity curves

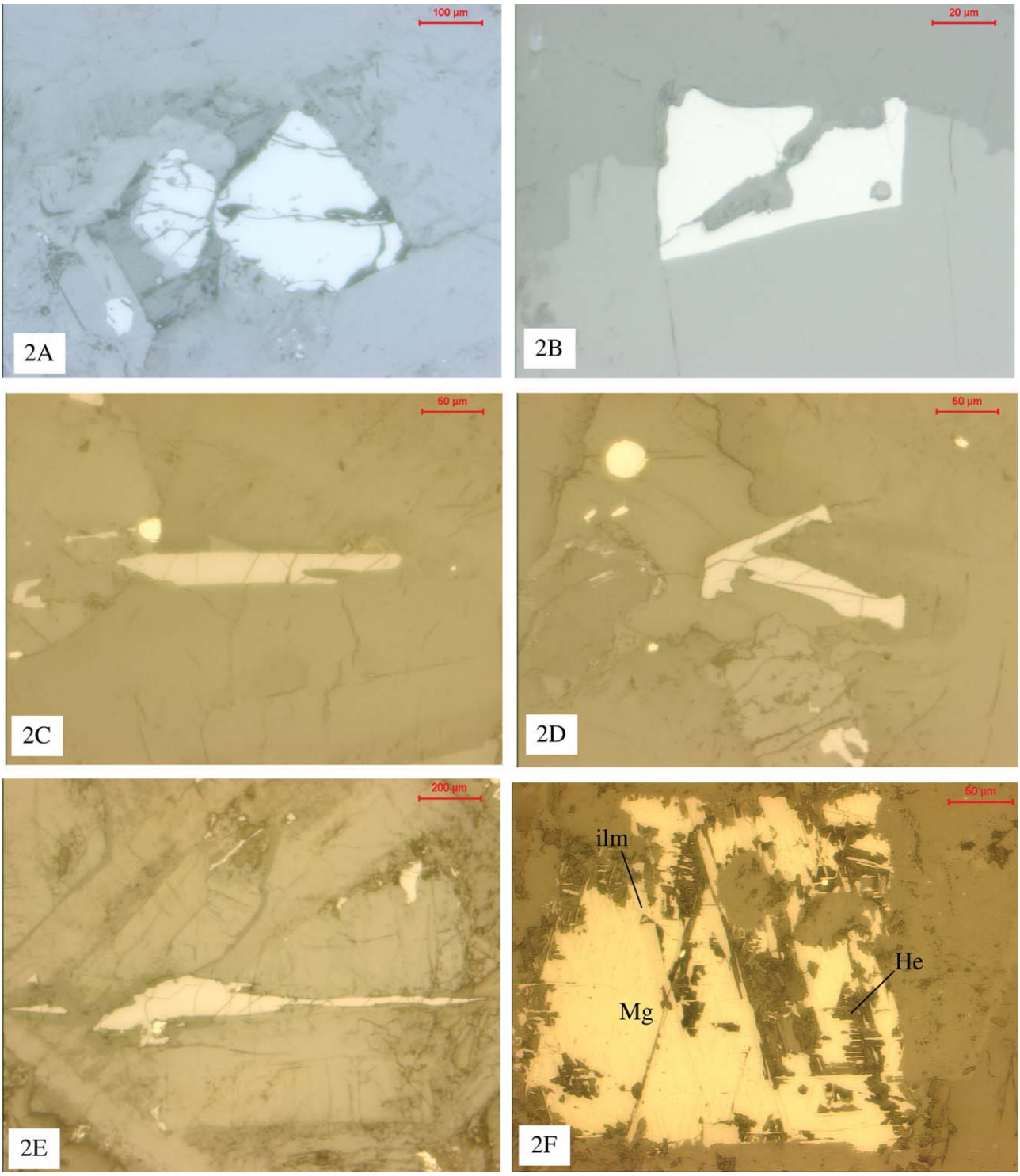


Plate 2. Photomicrograph of the basic rocks from the Nova Guarita dykes units. (A and B) (samples SD74O1 and SD75D1): unaltered crystals of magnetite; (C) (sample SD77-E1): hematite crystal; (D) (sample SD77-E1): martite in magnetite pseudomorph; (E) (sample SD77-E1): local intergranular and fracture growth of hematite; (F) (sample SD82-A1): small crystals of dark grey hematite (He) associated with magnetite (Mg) with ilmenite lamellae (ilm). All photos are with parallel polarizers.

are typical of magnetite or titanomagnetite mineral (mean destructive fields (MDF) between 2.5 mT and 17.5 mT), with intensities decreasing to nearly zero at fields below 100 mT (Figs. 3 and 7a). Although thermal treatment was effective in isolating the ChRM direction for only 33% of samples (Fig. 2), unblocking temperatures between 500 °C and 580 °C are responsible for about 50% up to almost 100% of the total intensity loss (Figs. 2 and 7b), which also suggests magnetite as the main magnetic carrier. These results match the microscopic observations which indicated a large fraction of magnetite among the opaque grains.

High-temperature thermomagnetic curves (low-field susceptibility versus temperature) show Curie temperatures around 580 °C and a pronounced Hopkinson peak (Fig. 8), typical of thermally stable SD/PSD magnetite grains (Dunlop and Özdemir, 1997). The curves are nearly reversible, although a small intensity fall at around 350–400 °C appears in some samples suggesting

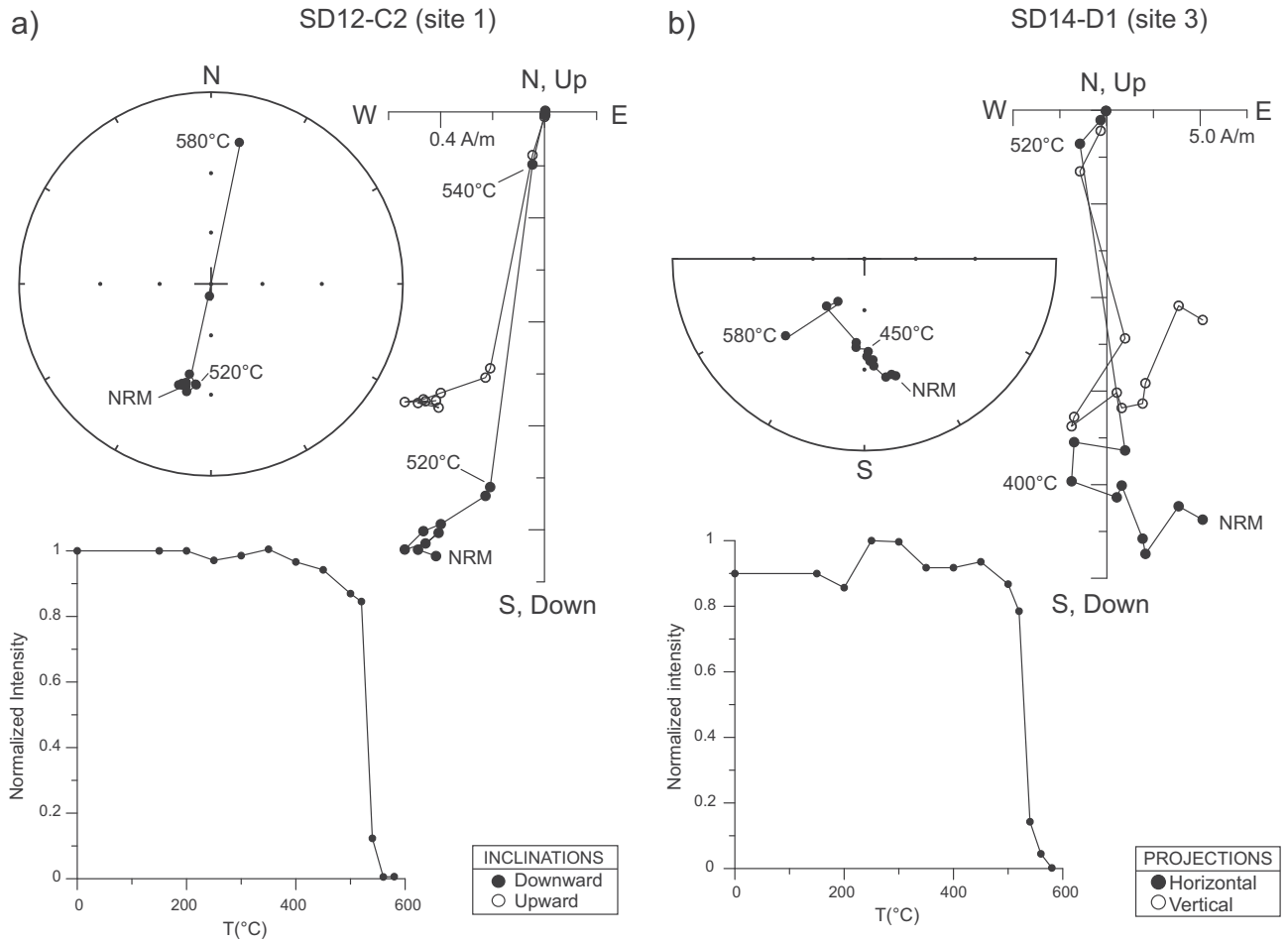


Fig. 2. Examples of thermal demagnetization. Figure shows Normalized intensity curves (M/M_o versus alternating field (H) curves), orthogonal projections and stereographic projections.

Table 1
Paleomagnetic data from Nova Guarita dykes.

Site	Localization	Samples	Site mean direction					VGP	
			n/N	Dec (°)	Inc (°)	α_{95} (°)	K	Plong (°E)	Plat (°N)
1	10.35°S/55.36°W	SD12	17/17	201.4	30.5	4.8	55.6	233.2	−68.3
2	10.48°S/55.22°W	SD13	13/12	231.9	42.6	10.0	19.6	238.1	−38.8
3	10.49°S/55.20°W	SD14	11/11	232.7	52.2	6.6	48.4	248.1	−36.7
4	09.25°S/57.59°W	SD58-60	13/9	41.3	−34.8	17.1	9.9	50.2	49.1
5	10.44°S/55.31°W	SD74	17/12	239.5	37.3	8.8	25.3	232.9	−32.1
6	10.42°S/55.29°W	SD75	12/10	228.5	47.7	10.6	21.6	244.0	−41.2
7	10.42°S/55.29°W	SD77	11/10	232.8	42.9	8.6	32.5	238.3	−38.0
8	10.42°S/55.29°W	SD78	14/13	231.3	57.7	7.2	33.9	255.2	−36.5
9	10.42°S/55.30°W	SD83	16/12	230.0	44.1	8.3	28.4	239.9	−40.4
10	10.42°S/55.30°W	SD84	14/13	220.8	37.3	9.2	21.4	234.6	−49.5
11	10.42°S/55.30°W	SD85	13/12	203.1	50.9	4.2	109.0	262.8	−59.9
12	10.42°S/55.30°W	SD86	13/12	214.8	39.5	5.9	55.9	238.9	−54.7
13	10.32°S/55.41°W	SD87	15/12	214.2	23.8	7.3	36.7	221.5	−56.5
14	10.32°S/55.42°W	SD89	12/11	218.9	29.4	10.2	20.9	226.9	−51.7
15	10.24°S/54.84°W	SD107	12/9	208.1	50.0	5.6	84.4	257.5	−56.8
16	10.08°S/55.12°W	SD113	16/12	17.2	−46.6	7.9	31.2	84.9	66.0
17	10.09°S/55.13°W	SD116	16/12	201.8	67.9	6.6	43.7	285.4	−45.4
18	10.11°S/55.11°W	SD117	16/9	259.2	62.2	5.7	81.9	257.4	−14.7
19	10.13°S/55.12°W	SD118	17/10	208.6	56.7	6.5	56.1	266.1	−52.6
Mean			19	220.5	45.9	6.5	27.7		
NG pole						7.0	23.7	245.9	−47.9

Notes: n/N – number of analyzed samples/number of samples used in the mean; Dec – declination; Inc – inclination; α_{95} – radius of the 95% confidence cone; K – precision parameter (Fisher, 1953); VPG – virtual geomagnetic pole; Plong – paleolongitude; Plat – paleolatitude.

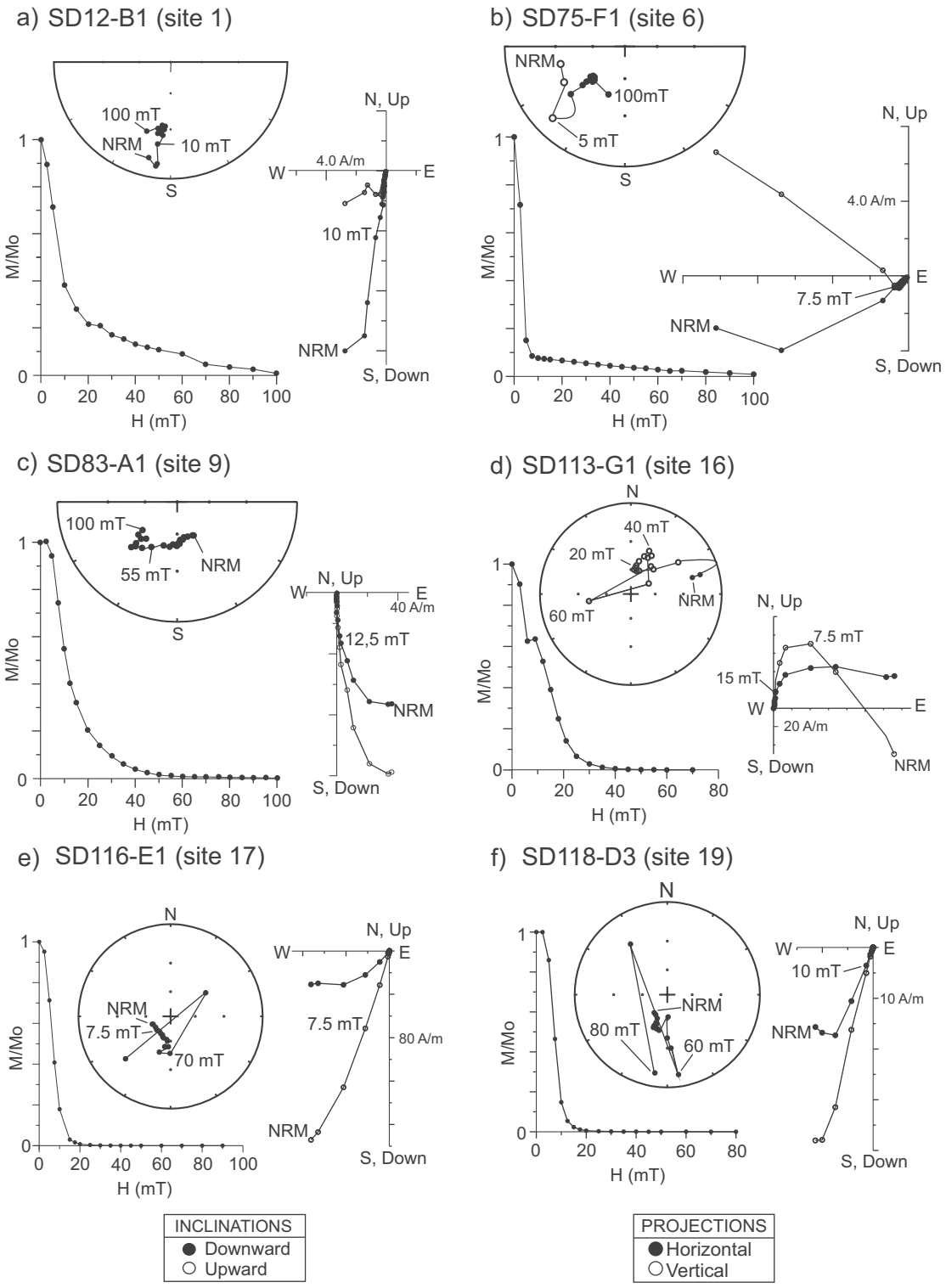


Fig. 3. Examples of alternating field demagnetizations and vector plots for samples from six sites. Figure shows Normalized intensity curves (M/M_o versus alternating field (H) curves), orthogonal projections and stereographic projections.

the presence of maghemite in small amounts (Fig. 8b and d). The well-characterized Verwey transition (−153 °C) observed in the low-temperature thermomagnetic curves (Fig. 8e and f) also corroborates the presence of magnetite as the main magnetic carrier.

Narrow-waist hysteresis curves were obtained for these samples, whose low-coercivities are typical of titanomagnetite or magnetite (Fig. 9). In Day's diagram (Day et al., 1977, modified after Dunlop, 2002), samples are concentrated in the pseudo-single domain (PSD) field for pure magnetite plotting along the SD–MD mixing line (Fig. 10), except for one sample that falls in the SP–SD field. The IRM curves are practically identical, reaching the saturation remanent magnetization in fields below 300 mT, consistent with a dominance of low-coercivity grains and reinforcing the interpretation of magnetite and titanomagnetite as the main remanence carriers in these rocks (Fig. 11).

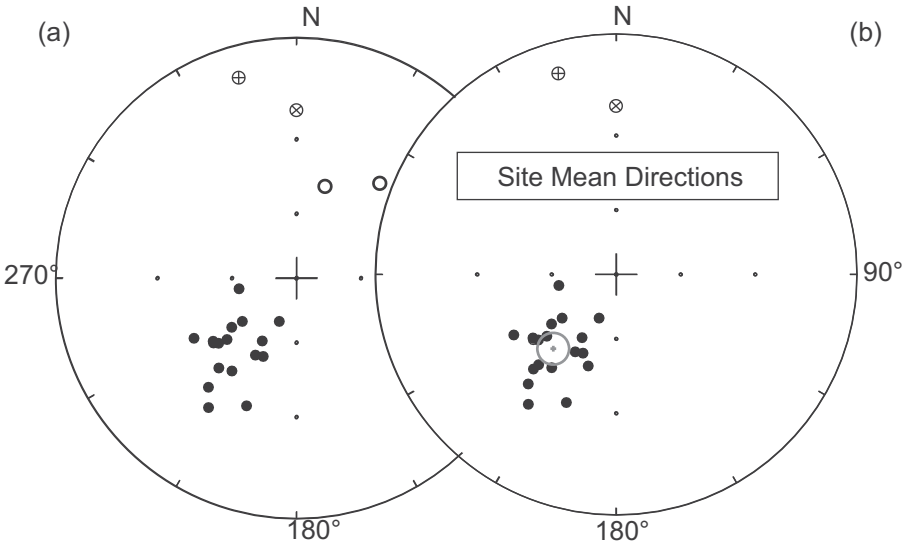


Fig. 4. Site mean directions: (a) normal and reverse directions, (b) after inversion of 'normal' polarity directions. Full (empty) circles represent downward (upward) inclinations. Symbols ⊗ and ⊕ are the present dipolar and geomagnetic fields, respectively.

7. Anisotropy of magnetic susceptibility (AMS)

Measurements of low-field anisotropy of magnetic susceptibility (AMS) were performed in 505 specimens of 19 sites belonging to the Nova Guarita dyke swarm. The directional and scalar data are presented in Table 2. The site mean magnetic susceptibility ($K_m = (K_1 + K_2 + K_3)/3$) ranges from 5.37×10^{-3} up to 9.27×10^{-2}, with an average of 7.05×10^{-2} (SI units). The high magnetic susceptibility values measured in these rocks are typical of unmetamorphosed basaltic rocks with a significant ferromagnetic contribution (Rochette, 1987).

Excluding the anomalous site 4 (Table 2), the degree of anisotropy ($P = K_1/K_3$) is lower than 10% (mean value of 1.044) suggesting that the characteristic remanent magnetization direction was not significantly affected by magnetic anisotropy (Hrouda, 1982; Cogné, 1988; Raposo et al., 2003; Salminen and Pesonen, 2007). In general, the shape parameter (Jelinek, 1981) data show a uniform distribution (Fig. 12), the foliation and lineation presenting the same mean value of 1.022 (Table 2).

The AMS axes for most of the dykes are well grouped and three types of magnetic fabrics could be defined, according to directions of the dykes in the field (Rochette et al., 1992): 'normal', 'intermediate' and 'inverse' fabrics. Unfortunately, for only 12 dykes, representing 63% of the analyzed dykes, the dyke trends could be identified in the field. Thus, the 'normal' fabric was found in 75% (nine), 'intermediate' fabric in 8% (one) and 'inverse' fabric was observed in 17% (two) of these dykes (Fig. 13).

According to Rochette et al. (1992), the 'intermediate' fabric can be explained by the presence of fine-grained, particularly SD–PSD grains. Indeed, magnetic mineralogy experiments (see above) might be interpreted as PSD-magnetite to be the main magnetic carrier of almost all dykes. Alternatively, the intermediate fabric can be assigned to a vertical compaction of a static magma column with minimal stress along the direction of the dyke

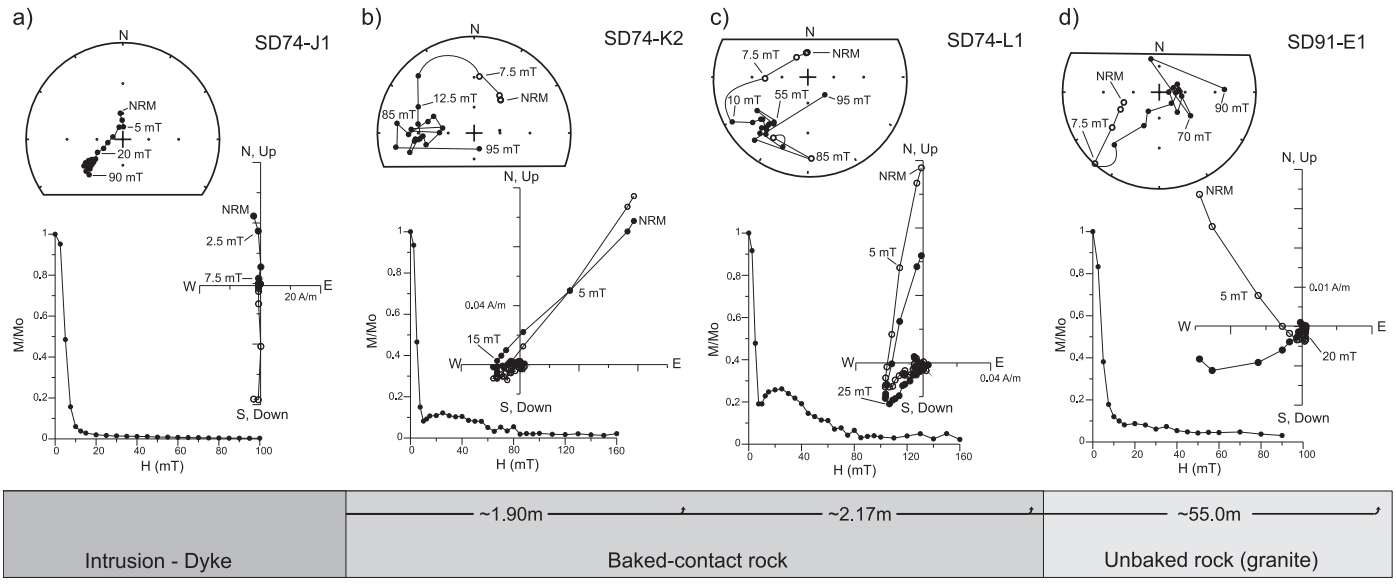


Fig. 5. Baked-contact test: AF demagnetization of a sample from the dyke (a), of a granite (baked-contact) at ~1.90 m from intrusion (b), of a granite (baked-contact) at ~2.17 m from intrusion (c), and of a granite (unbaked) at ~55 m from intrusion (d).

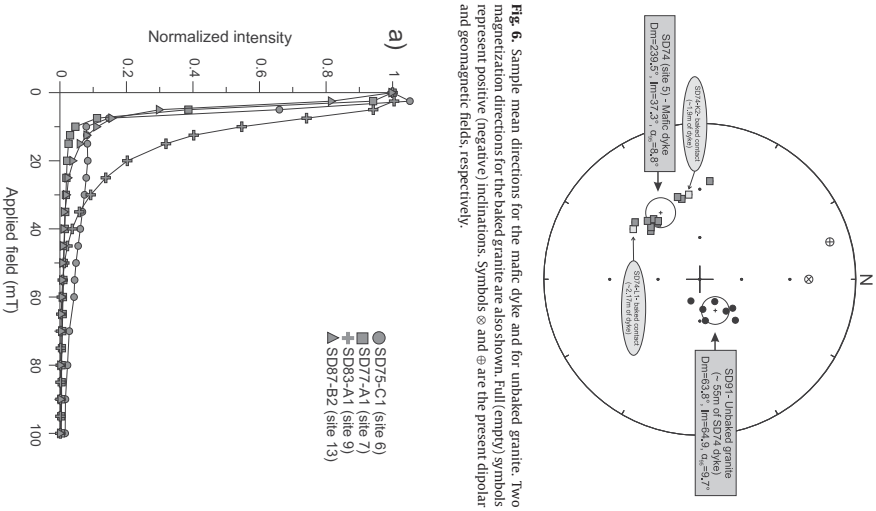


Fig. 6. Sample mean directions for the mafic dyke and for unbaked granite. Two magnetization directions for the baked granite are also shown. Full (empty) symbols represent positive (negative) inclinations. Symbols ⊗ and ⊕ are the present dipolar and geomagnetic fields, respectively.

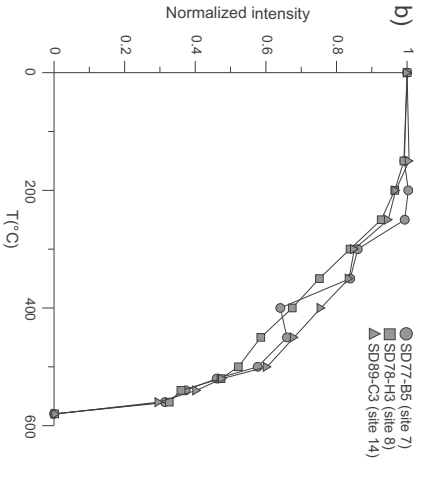


Fig. 7. Examples of normalized magnetization intensities during (a) alternating magnetic field and (b) thermal demagnetizations for samples from different sites.

Table 2
AMS data.

Site	Strike	Samples	*N*	K_m	*L*	*F*	*P*	*T*	K_1			K_2			K_3		
									Dec (°)	Inc (°)	*e*/*z*	Dec (°)	Inc (°)	*e*/*z*	Dec (°)	Inc (°)	*e*/*z*
1	N20E	SD12	17	3.74E−02	1.053	1.040	1.096	−0.338	56	64	13/3	207	23	13/6	302	11	6/3
2	N40W	SD13	15	9.22E−02	1.025	1.012	1.037	−0.365	37	37	18/4	138	13	49/14	244	50	49/12
3	N40W	SD14	05	9.27E−02	1.017	1.020	1.037	0.065	251	9	23/4	139	66	20/17	345	21	19/16
4	–	SD58-60	13	5.37E−03	1.132	1.172	1.326	0.160	276	36	11/4	171	20	11/4	58	48	10/6
5	N20E	SD74	62	6.92E−02	1.033	1.020	1.054	−0.311	65	44	26/12	191	34	26/22	305	27	24/11
6	N40W	SD75	47	8.81E−02	1.028	1.030	1.058	0.023	108	7	23/7	17	7	27/14	245	80	25/8
7	N40W	SD77	29	8.78E−02	1.011	1.014	1.025	0.056	62	25	24/20	156	8	33/24	263	63	34/18
8	N30W	SD78	11	8.80E−02	1.011	1.011	1.023	0.025	295	77	9/7	178	6	14/9	87	11	14/6
9	–	SD83	27	8.49E−02	1.012	1.023	1.036	0.254	145	45	20/18	35	19	19/12	289	39	21/5
10	–	SD84	43	8.80E−02	1.015	1.011	1.011	−0.145	1	7	11/7	260	57	53/10	96	32	53/7
11	–	SD85	27	8.48E−02	1.009	1.012	1.022	0.129	218	5	18/7	125	25	19/11	319	64	11/7
12	E–W	SD86	36	7.94E−02	1.011	1.017	1.028	0.202	165	9	35/10	257	10	35/7	33	76	10/8
13	–	SD87	24	5.50E−02	1.023	1.018	1.041	−0.122	307	3	17/12	209	71	58/15	38	18	58/12
14	–	SD89	21	5.89E−02	1.013	1.017	1.030	0.042	336	17	26/9	77	31	26/19	221	53	20/9
15	–	SD107	42	3.25E−02	1.007	1.011	1.019	0.116	54	49	27/10	243	40	23/11	149	4	12/10
16	N15W	SD113	35	7.13E−02	1.010	1.007	1.018	−0.230	196	9	18/12	106	2	28/17	1	80	28/11
17	N5W	SD116	16	6.30E−02	1.027	1.020	1.047	−0.144	171	26	7/2	320	60	7/4	75	13	4/2
18	N5W	SD117	27	7.80E−02	1.017	1.017	1.034	0.013	200	8	22/6	108	20	17/11	311	68	19/13
19	N45E	SD118	08	4.43E−02	1.007	1.045	1.053	0.713	236	20	47/3	351	49	47/4	132	35	4/4
		Mean	505	7.05E−02	1.022	1.022	1.044	−0.030									

Notes: *N* – number of measured specimens; $K_m = (K_1 + K_2 + K_3)/3$ – bulk mean magnetic susceptibility; $L = K_1/K_2$ – magnetic lineation; $F = K_2/K_3$ – magnetic foliation, $P = K_1/K_3$ – degree of anisotropy; $T = [2\ln(K_2/K_3)/\ln(K_1/K_3)] - 1$ – Jelinek's shape parameter (Jelinek, 1981); Dec – declination; Inc – inclination; K_1, K_2 and K_3 are mean AMS eigenvectors, which represent the maximum, intermediate and minimum susceptibility-axes, respectively; *e*/*z* – semi-angles of the major and minor axes of the 95% confidence ellipse, respectively, calculated by the bootstrap method.

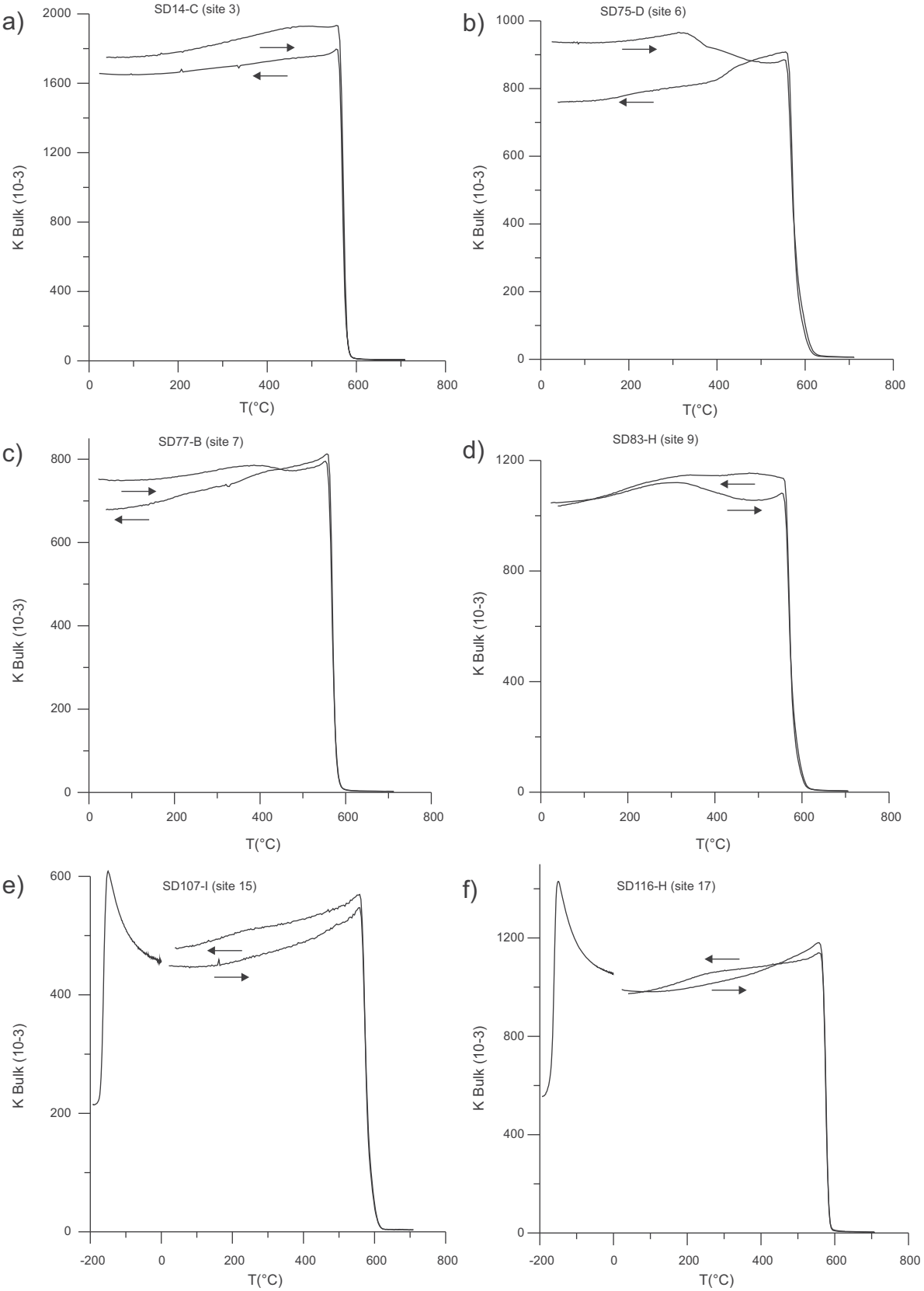


Fig. 8. Thermomagnetic curves showing variation in magnetic susceptibility (SI) with low and high temperature. Curves were corrected from furnace effects.

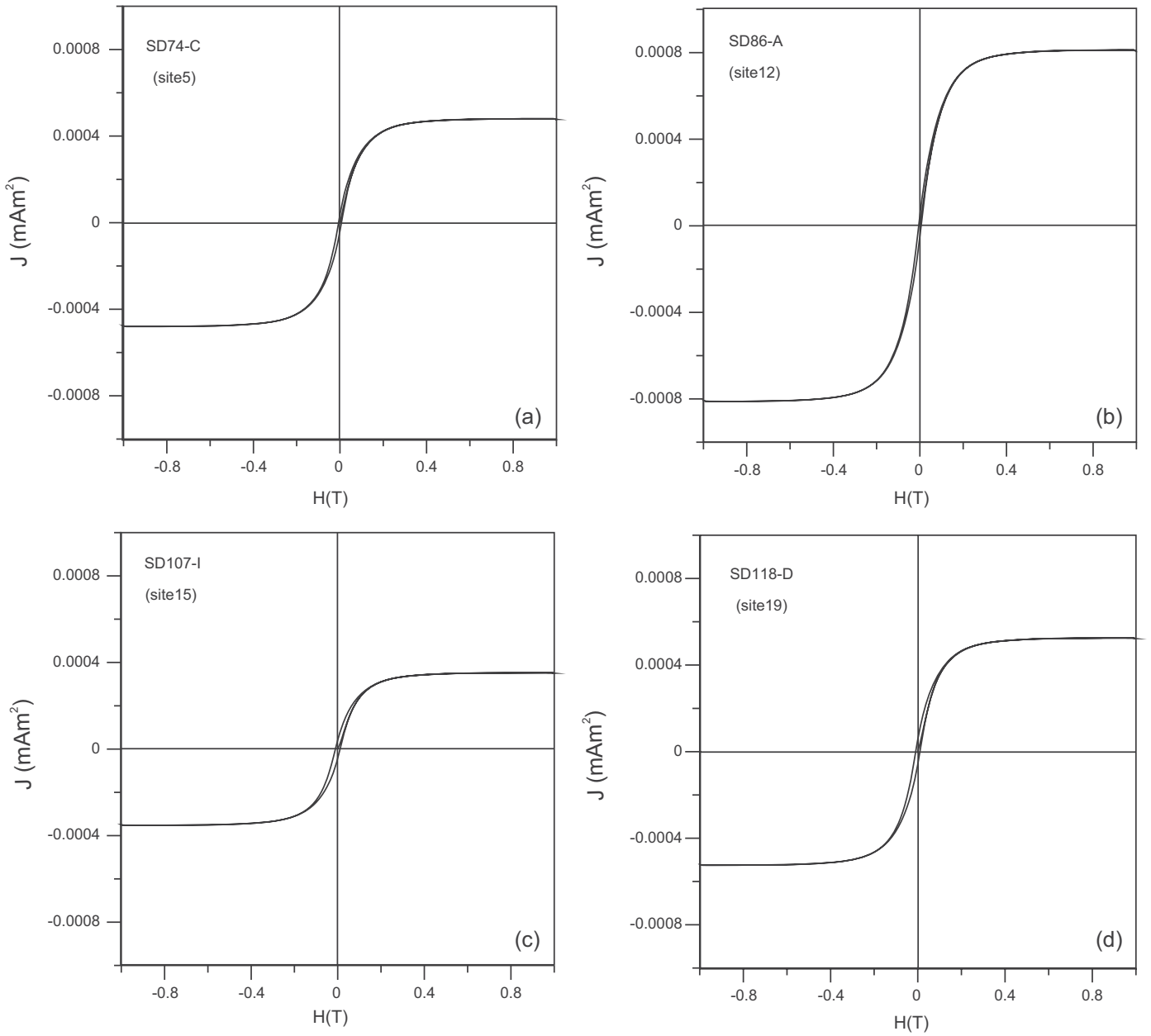


Fig. 9. Hysteresis curves showing narrow waist type indicating the presence of titanomagnetite or magnetite.

(Park et al., 1988; Raposo and D'Agrella-Filho, 2000) or caused by stress thermally controlled (Ellwood, 1979). The inverse fabric is normally interpreted as a result of secondary processes such as hydrothermal alteration, post-emplacement modification, or presence of SD grains in the dykes investigated (Rochette et al., 1992). Dykes with 'normal' fabric are usually interpreted as a flow fabric and K_1 is used to infer the direction of magma flow (Knight and Walker, 1988). The K_1 inclination indicates that 63% of the dykes were fed by a horizontal or sub-horizontal magma flow ($I_1 < 30^\circ$), 26% by inclined flow ($30^\circ < I_1 < 60^\circ$), and only 11% by a vertical flow ($I_1 > 60^\circ$). These results imply that most dikes were fed by a distant source (Knight and Walker, 1988; Raposo and Ernesto, 1995; Raposo and D'Agrella-Filho, 2000; Raposo and Berquó, 2008).

8. ^{40}Ar–^{39}Ar geochronology

Since no radiometric ages are presently available for the studied dykes we selected samples from four sites (sites 6, 7, 8, and 9, Table 1) for ^{40}Ar/^{39}Ar dating. Selection was based on petrographic analysis considering two criteria: the best preserved igneous texture and the less altered as possible samples. Biotite and plagioclase grains were separated for dating employing the methods described in the legend of Table 1 (Supplementary data file). After a decay period following irradiation, samples were analyzed by laser ^{40}Ar/^{39}Ar heating according to the procedures detailed in Vasconcelos et al. (2002). Results are shown in Table 1 (Supplementary data file). The best ^{40}Ar–^{39}Ar results were obtained for biotite grains (Fig. 14). All but one ^{40}Ar–^{39}Ar step-heating spectra define very consistent plateau ages (as defined by Fleck et al., 1977, see legend of the Supplementary data file of Table 1) between 1407 ± 8 Ma and 1430 ± 8 Ma. The aliquot 6523-01 from sample SD77-G1 (Fig. 14) yielded a pseudo-plateau age of 1403 ± 8 Ma where more than three steps are within the 2σ criterion for a plateau but the steps correspond to less than 50% of the total ^{39}Ar released. Ideograms (age-probability diagrams) were constructed for all analyzed samples (Fig. 14) yielding weighted mean ages of

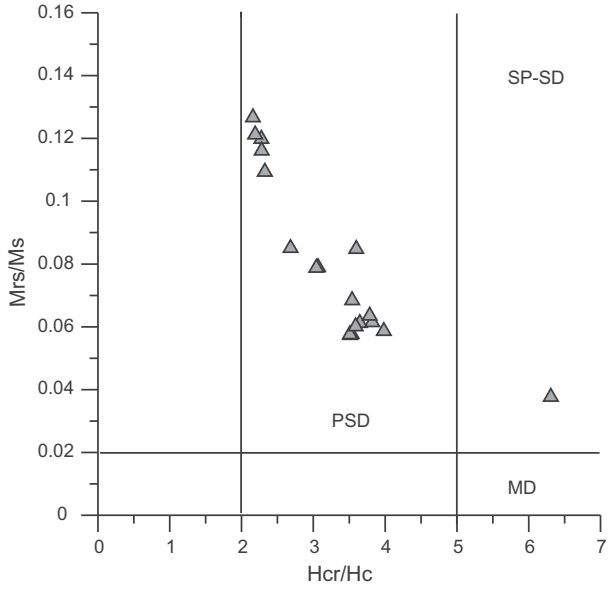


Fig. 10. Day's diagram (1977), after Dunlop (2002), indicating that most samples fall into pseudo-single domain (PSD) field. M_s = saturation magnetization; M_{rs} = saturation isothermal remanence; H_{cr} = coercivity of remanence; H_c = bulk coercive force.

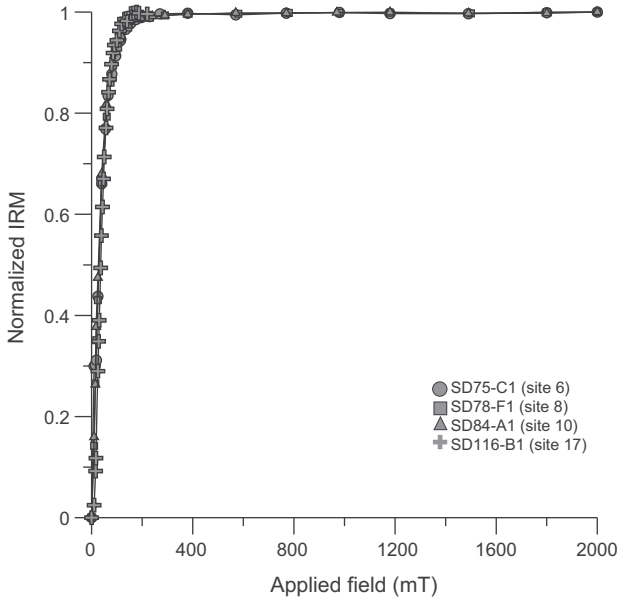


Fig. 11. IRM acquisition curves (normalized intensities versus magnetic field).

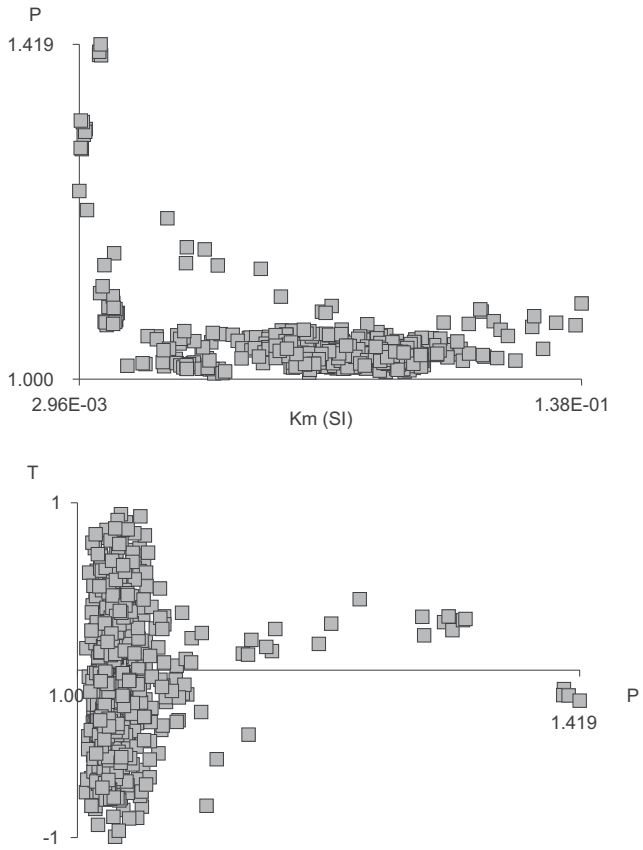


Fig. 12. (a) Degree of anisotropy (P) versus bulk susceptibility ($K_m = (K_1 + K_2 + K_3)/3$) for all analyzed samples; (b) $T = [2\ln(K_2/K_3)/\ln(K_1/K_3)] - 1$ – Jelinek's shape parameter (Jelinek, 1981) versus degree of anisotropy ($P = K_1/K_3$) for the analyzed samples.

1416 ± 13 Ma for sample SD75-E1 (site 6), 1406 ± 5 Ma for sample SD77-G1 (site 7), 1418 ± 16 Ma for sample SD78-A5 (site 8), and 1424 ± 16 Ma for sample SD83-A2 (site 9), which are interpreted as the most correct estimates for the ages of biotites in these samples.

^{40}Ar–^{39}Ar dating in plagioclase grains yielded worst results (Fig. 15). Neither of the aliquots from samples SD75-E2 and SD83-A2 produced plateau ages according to the definition of Fleck et al. (1977). The step heating profiles are perturbed by variable ^{40}Ar and/or K loss most likely due to alteration of the plagioclase and/or ^{40}Ar loss due to post-crystallization thermal effects and no useful age can be determined. The most useful geochronological information is that they were partially re-set (by a heating

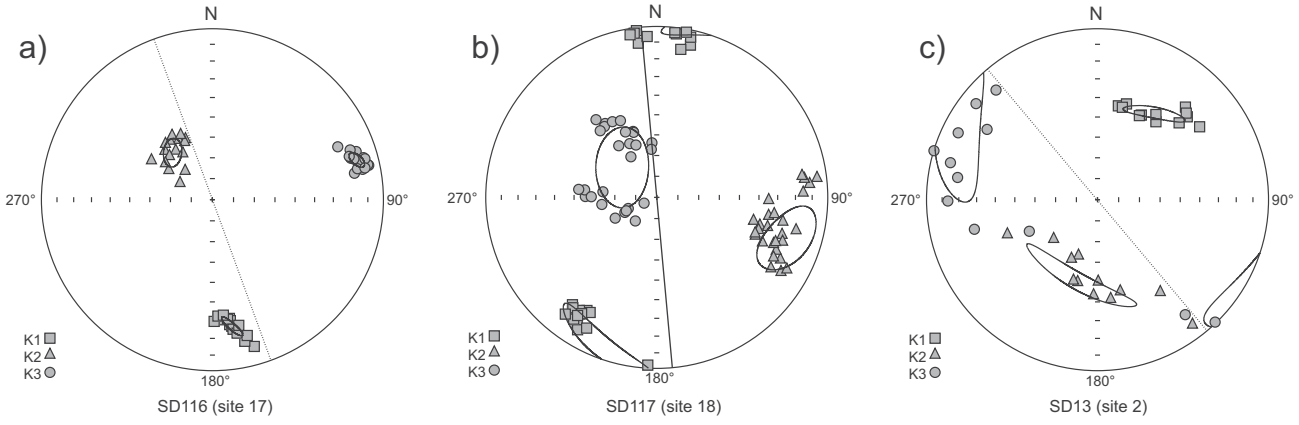


Fig. 13. Stereoplots of directions of susceptibility axes, which indicate AMS fabrics: (a) 'normal' fabric, (b) 'intermediate' fabric and 'inverse' fabric. Symbols: K_1 (square), K_2 (triangle) and K_3 (circle) are maximum, intermediate and minimum AMS axes, respectively. The hatched line denotes the orientation of the dyke plane.

event) at ca. 1000 Ma or that they only cooled below the closure temperature of plagioclase around that time. Only aliquot 6539-02 from sample SD77-A2 (site 7, Fig. 15) produced a plateau age (1380 ± 17 Ma) according to the definition of Fleck et al. (1977). The step heating profiles for both aliquots, however, are perturbed most likely due to a common thermal event subsequent to plagioclase crystallization. But the highest temperature steps from both profiles (1392 ± 11 Ma, 1412 ± 23 Ma, 1397 ± 23 Ma, 1403 ± 30 Ma,

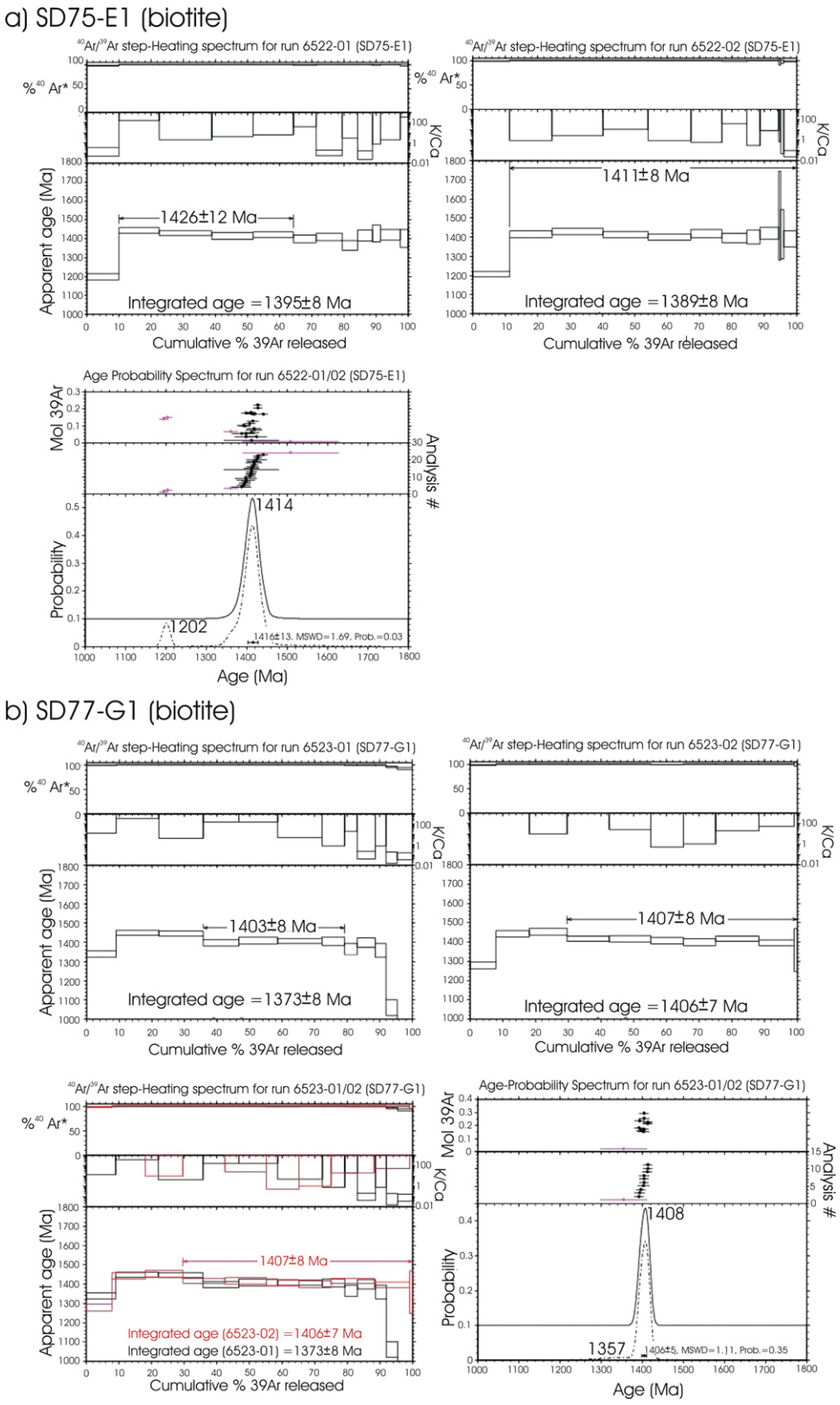


Fig. 14. Apparent ages (Ma) as a function of cumulative ^{39}Ar released for biotite separate (two runs) from samples SD75-E1 (a), SD77-G1 (b), SD78-A5 (c), and SD83-A2 (d). Age Probability Spectra for all samples are shown (see text for details).

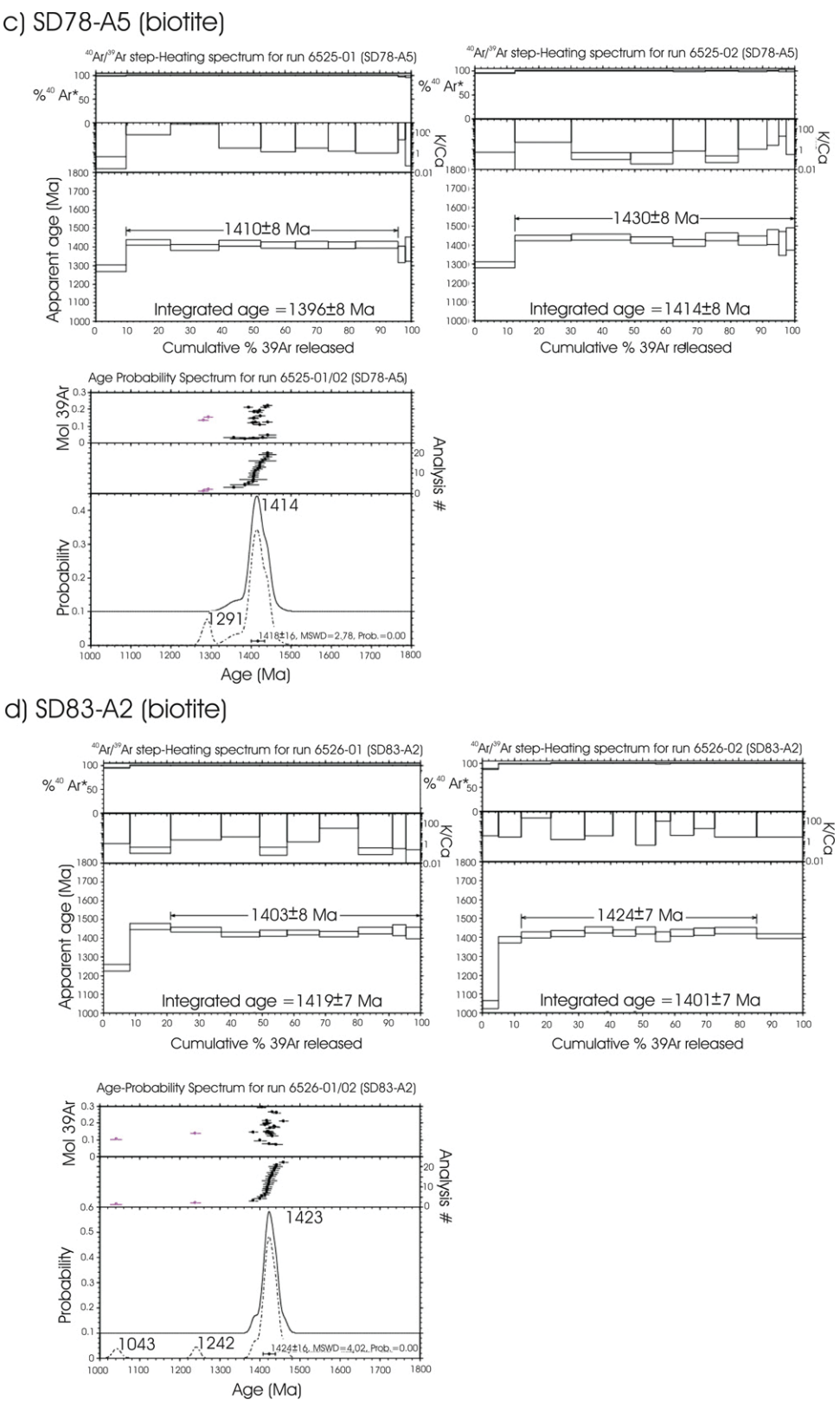


Fig. 14.

Table 1 Supplementary data file) are within error of the age determined for the coexisting biotite in the rock (Fig. 14). The highest temperature probability peak of 1392 Ma on the age probability diagram (Fig. 15) is also close to the biotite age. Due to the thermal age resetting of the sample, the plateau age of 1380 ± 17 Ma should be regarded as a minimal age for this sample. Concerning sample SD78-A3 (site 8) both plagioclase aliquots produced plateau ages, according to the definition of Fleck et al. (1977), although the plateau ages are very different (1421 ± 13 Ma, 1202 ± 14 Ma) and reflect different events. However, the highest temperature plateau

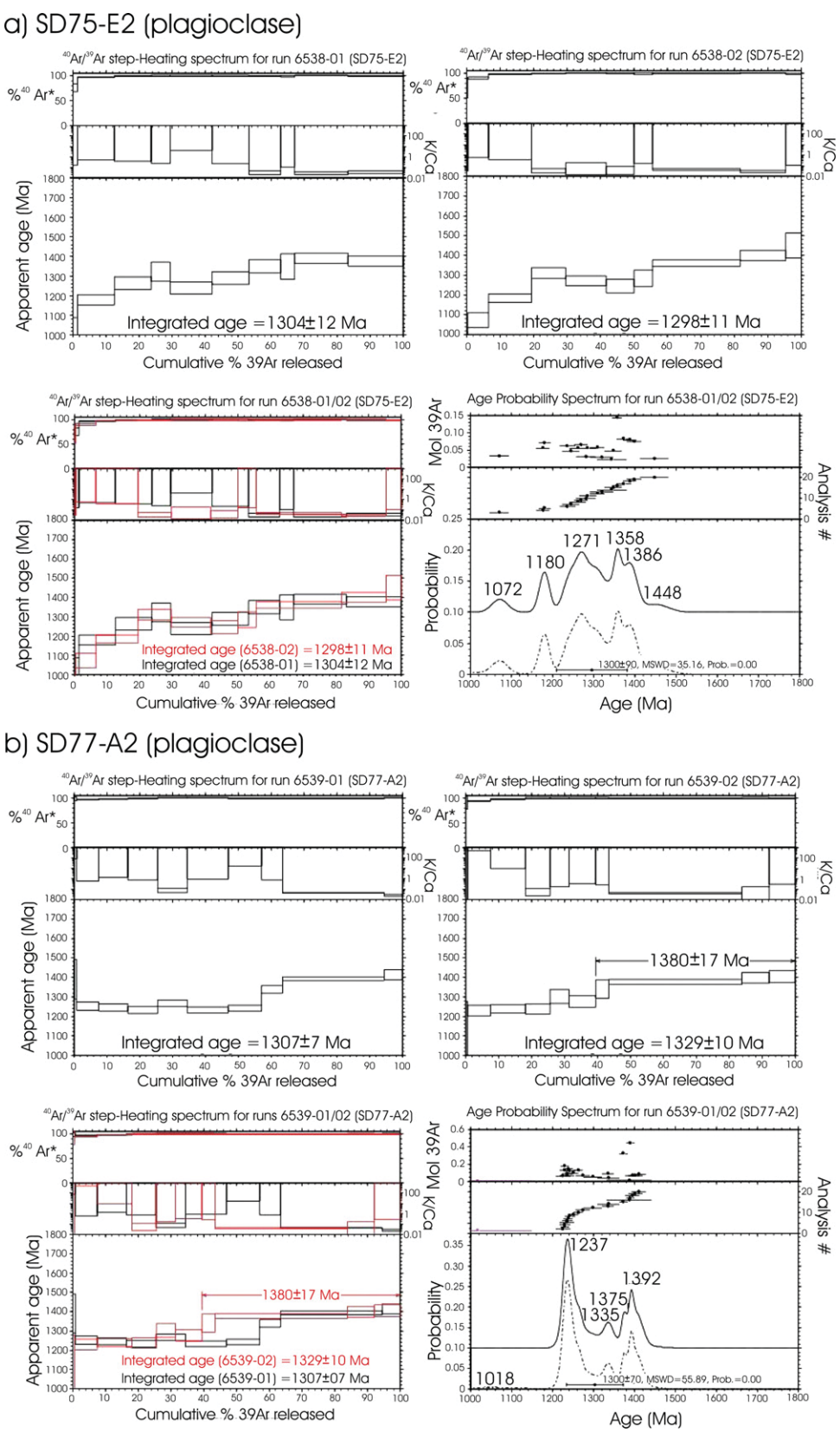


Fig. 15. Apparent ages (Ma) as a function of cumulative ^{39}Ar released for plagioclase separate (two runs) from samples SD75-E2 (a), SD77-A2 (b), SD78-A3 (c), and SD83-A2 (d). Age Probability Spectra for all samples are shown (see text for details).

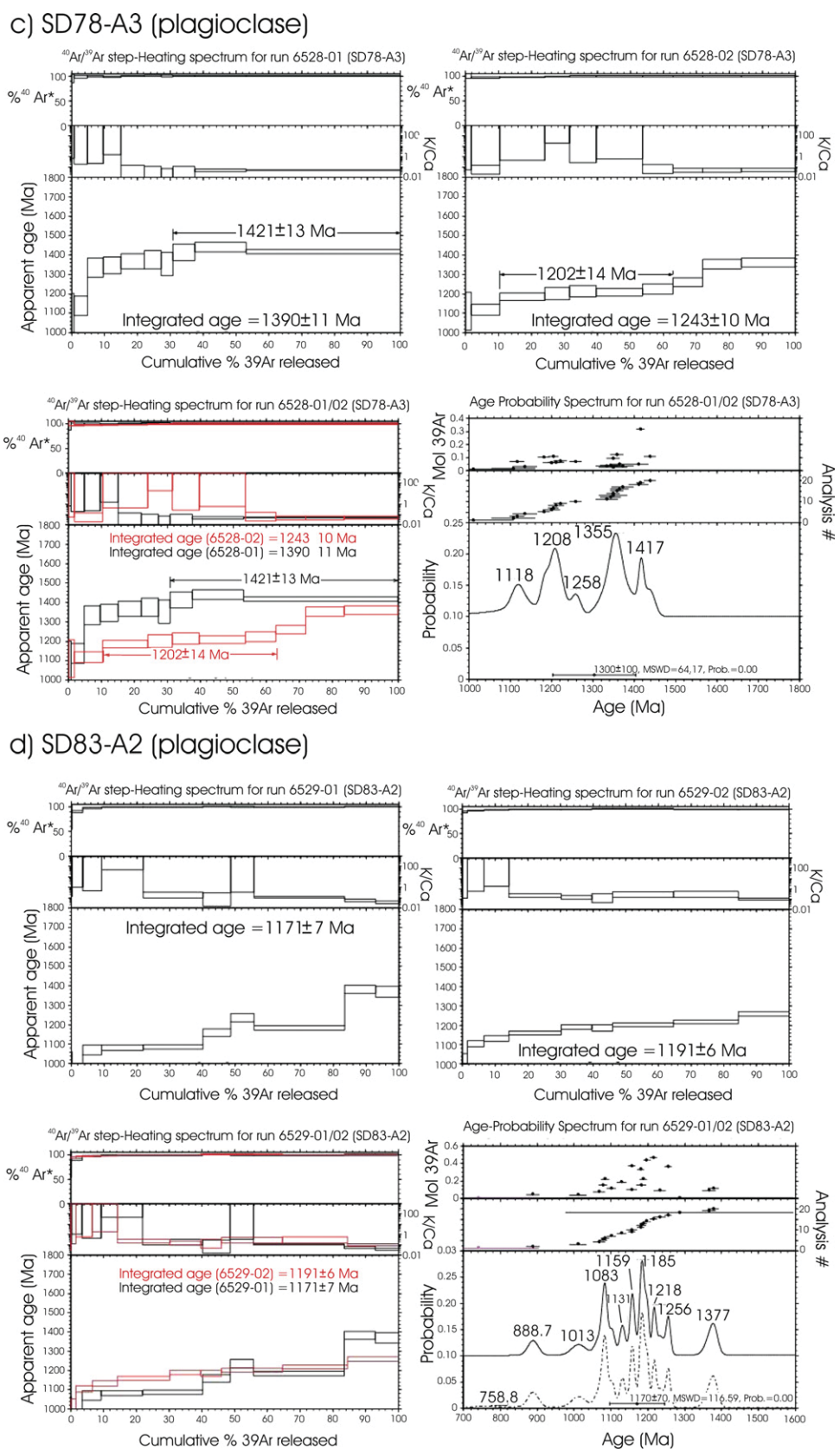

c) SD78-A3 (plagioclase)
40Ar/39Ar step-Heating spectrum for run 6528-01 (SD78-A3)
40Ar/39Ar step-Heating spectrum for run 6528-02 (SD78-A3)
%40 Ar*
K/Ca
Apparent age (Ma)
1421±13 Ma
Integrated age =1390±11 Ma
1202±14 Ma
Integrated age =1243±10 Ma
Cumulative % 39Ar released
40Ar/39Ar step-Heating spectrum for run 6528-01/02 (SD78-A3)
Integrated age (6528-02) =1243 10 Ma
Integrated age (6528-01) =1390 11 Ma
Age Probability Spectrum for run 6528-01/02 (SD78-A3)
Mol 39Ar
Analysis #
Probability
1208
1355
1417
1118
1258
1300±100, MSWD=64,17, Prob.=0.00
Age (Ma)
d) SD83-A2 (plagioclase)
40Ar/39Ar step-Heating spectrum for run 6529-01 (SD83-A2)
40Ar/39Ar step-Heating spectrum for run 6529-02 (SD83-A2)
Integrated age =1171±7 Ma
Integrated age =1191±6 Ma
40Ar/39Ar step-Heating spectrum for run 6529-01/02 (SD83-A2)
Integrated age (6529-02) =1191±6 Ma
Integrated age (6529-01) =1171±7 Ma
Age-Probability Spectrum for run 6529-01/02 (SD83-A2)
1159
1185
1083
1131
1218
1256
1377
888.7
1013
758.8
1170±70, MSWD=116.59, Prob.=0.00

Fig. 15.

age (1421 ± 13 Ma) is within error of the age determined for the coexisting biotite (Fig. 14) and corresponds to the crystallization age of plagioclase in this rock. Here again, the step heating profiles for both aliquots are perturbed due to thermal event subsequent to plagioclase formation.

The very consistent ^{40}Ar–^{39}Ar results described above indicates a Mesoproterozoic age for this magmatic event. Taking into account only the ^{40}Ar–^{39}Ar plateau ages in biotite (Fig. 14), a mean age of 1418.5 ± 3.5 Ma may be calculated, which is interpreted as the best estimate age of the Nova Guarita mafic intrusives.

9. Discussion

Both 'reversed' (south/southwestern, 17 sites) and 'normal' (northeastern, 2 sites) stable directions (NG component) were disclosed in samples from the Nova Guarita mafic intrusives. Magnetic experiments indicate that the NG component is carried by pseudo-single domain magnetite grains with high-H_c and high-unblocking temperatures, denoting its high magnetic stability. The petrographically observed high-temperature oxidation/exsolution upon the initial cooling of the magma may be responsible for this high stability of magnetite (Hodych, 1996; Dunlop and Özdemir, 1997; Haggerty, 1981). A positive baked contact test obtained for site 5 (see above) confirms its primary origin and implies that the NG component is a thermo-remanent magnetization acquired during rock cooling at about 1415–1422 Ma ago.

The Nova Guarita pole (NG: 245.9°E, 47.9°S, N = 19, A_{95} = 7.0°, K = 23.7) satisfies 6 out of the 7 quality criteria proposed by Van der Voo (1990) as follows: (1) ^{40}Ar–^{39}Ar dating on biotites from four dykes yielded well-defined and concordant plateau ages, with a mean of 1418.5 ± 3.5 Ma, which likely represent the age of magnetization acquisition; (2) the NG pole was derived from a set of 218 specimens (19 sites) and has adequate Fisher's statistical parameters (A_{95} = 7.0°, K = 24); (3) remanence vectors were isolated by least-squares fit from orthogonal diagrams (Kirschvink, 1980) after stepwise AF demagnetization; (4) positive baked contact test performed in one dyke confirm the primary thermoremanent nature of the magnetization; (5) the dykes intrude Paleoproterozoic granites from the southernmost part of the 1950–1800 Ma Ventuari-Tapajós Province, and there are no geological evidence of later tectonic events in the studied area; (6) both reverse and normal polarities were recorded, attesting that a long-time has elapsed during dykes intrusions, enough to eliminate the secular variation of the geomagnetic field. Criterion 7 is the only not satisfied since similar directions were found for 980 Ma mafic sills from the Rio Branco area and 1150 Ma sediments from the Aguapeí Group (Elming et al., 2009; D'Agrella-Filho et al., 2008). However, this area is approximately 600 km to the southwest of our studied area, and with reference to the positive outcome of the baked contact test it is very unlikely that the mafic sills remagnetized the studied dykes. Someone may argue that the Ar–Ar data revealed a possible thermal event that affected partially the plagioclase isotopic system at about 1000 Ma ago. But, the low closure temperatures associated with plagioclase (~200 °C) are not enough to completely erase the high-blocking temperatures associated with magnetization in the Nova Guarita dykes (Pullaiah et al., 1975).

In sum, the Nova Guarita pole (NG) may be regarded as a key pole (Q = 6), and can be used to constrain the paleogeographic position of the proto-Amazonian Craton at ~1419 Ma, and test the longevity of the Nuna supercontinent.

9.1. Nuna supercontinent

The existence of a great continental mass (Nuna supercontinent) at around 1780 Ma ago comprised by Laurentia, Baltica, North China and Amazonian Craton has been proposed by Bispo-Santos et al. (2008) based on selected key paleomagnetic poles. This configuration, slightly modified from the original one, is reproduced in Fig. 16a. We have used the following rotation Euler poles: 0.3°N, 7.2°E, −83.4° for Laurentia; 27.8°N, 163.2°E, 56.3° for Baltica; 39.8°N, 310.5°E, −68.1° for North China Craton; and 77.4°N, 93.0°E, 128.9° for Amazonian Craton. The first two rotation poles constrain

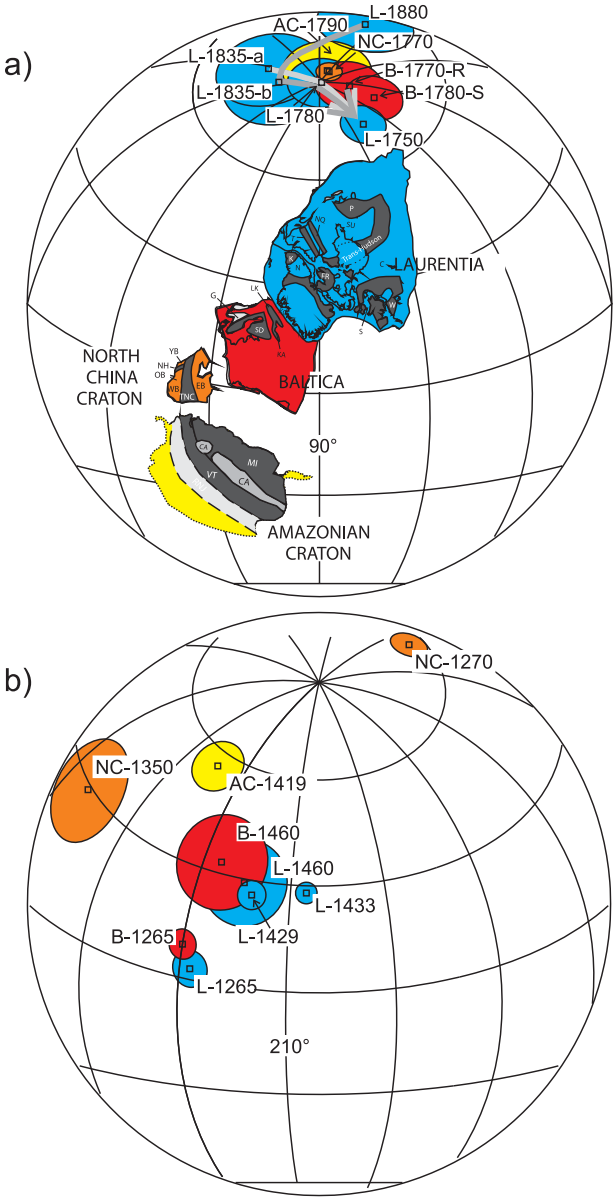


Fig. 16. (a) Nuna reconstruction at ~1780 Ma (modified after Bispo-Santos et al., 2008). Paleomagnetic poles (Table 3) were rotated using the same Euler rotation poles assigned for each continent in the reconstruction of Nuna. See text for details. Archaean cratonic areas (light grey): Laurentia (S – Slave; C – Churchill; SU – Superior; N – Nain and equivalent in southern Greenland), Baltica (KO – Kola; KA – Karelia), North China (YB – Yinshan Block; OB – Ordos Block; WB – West Block; EB – East Block), Amazonia (CA – Central Amazonia Province). Paleoproterozoic belts (dark grey): Laurentia (NQ – New Quebec; T – Torngat; W – Wopmay; P – Penokean; K – Kefilidian; NA – Nagssugtoqidian; FR – Foxe-Rinklan), Baltica (LK – Lapland-Kola; SD – Svecofennian Domain; G – Gothian Province), North China (NH – North Hebei/Khondolite belt; TNC – Trans-North China), Amazonia (MI – Maroni-Itacaiunas; VT – Ventuari-Tapajós; RNJ – Rio Negro-Juruena). (b) Selected paleomagnetic poles for Laurentia, Baltica, North China Craton and Amazonian Craton with ages between 1460 Ma and 1265 Ma after rotate them using the Euler poles – Laurentia (0.3°N, 7.2°E, −83.4°); Baltica (27.8°N; 163.2°E, 56.3°); North China Craton (39.8°N, 310.5°, −68.1°) and Amazonian Craton (77.0°N, 89.5°E, 122.1°).

Laurentia and Baltica in a similar configuration as that proposed by Salminen and Pesonen (2007).

The ~1770–1790 Ma poles (Table 3) selected by Bispo-Santos et al. (2008) for these four blocks are plotted in Fig. 16a after rotating them using the same rotation poles described above. Note that they cluster around the geographic pole for that time. Only one key paleomagnetic pole (reliability factor $Q = 5$) is available for the Amazonian Craton (AC-1790 in Fig. 16a), which was obtained for the well-dated, 1789 ± 7 Ma (U–Pb, zircon) unmetamorphosed felsic volcanic rocks from the Colider Suite (Bispo-Santos et al., 2008). Paleomagnetic analysis reveals a dual-polarity characteristic magnetic component interpreted to be a primary thermochemical remanent magnetization acquired during rock cooling.

A 1770 Ma key paleomagnetic pole is available for North China (NC-1770 in Fig. 16a). It was obtained for the well dated 1769.1 ± 2.5 Ma (U–Pb dating) Taihang dyke swarm (Halls et al., 2000) for which a single-'normal' polarity, northeastern component with shallow inclination was disclosed. This component passes baked contact tests indicating they represent a primary magnetization acquired during cooling of the dykes. Recently, a new paleomagnetic study of these dykes revealed a similar paleomagnetic pole corroborating the preliminary analysis (Hou et al., 2009).

Two reliable ca. 1770 Ma poles are available for Baltica (B-1770-R and B-1780-S in Fig. 16a), one of which was obtained for gabbros and dolerites from the well-dated 1770 ± 12 Ma (U–Pb) Ropruchey sill. Only a single 'normal' polarity was disclosed for these rocks, whose magnetization was regarded as primary (Fedotova et al., 1999). The other pole was obtained for sandstones from the Shosksha Formation. Paleomagnetic analysis reveals dual-polarity stable directions carried by early diagenetic, single domain (SD) hematite. Older granites dated at 1794 ± 24 and 1778 ± 16 Ma (U–Pb in zircon), and the 1770 Ma Ropruchey sill intrusive in the sediments constrain the age of the Shosksha Formation to between 1790 and 1770 Ma (Pisarevsky and Sokolov, 2001).

The most complicated situation is for Laurentia, partly due to the lack of key paleomagnetic poles in the 1770–1790 Ma time interval (Irving et al., 2004). Also, ambiguous interpretations have been proposed for the Paleoproterozoic paleolatitude of Laurentia, and the geodynamics of its formation along the Trans-Hudson mobile belt between 1900 and 1800 Ma (see Pesonen et al., 2003). The 1880–1830 Ma paleomagnetic poles from Trans-Hudson Orogenic Province (THO) and Superior craton yield high-paleolatitudes for Laurentia (e.g., Symons and Harris, 2005 and references herein), while syn-Hudsonian 1880–1830 Ma geological units from the western Slave–Rae–Hearne craton provided paleomagnetic poles that indicate moderate paleolatitudes for Laurentia (e.g., Irving et al., 2004 and references herein). In their reconstruction, Bispo-Santos et al. (2008) have chosen the paleomagnetic pole (L-1780 in Fig. 16a) obtained for the syn-Hudsonian Dubawnt Group (Park et al., 1973) from Rae Province (Western Laurentia), whose age is constrained to be between 1830 and 1760 Ma (Rainbird and Hadlari, 2000). Results obtained for sedimentary and igneous rocks from this Group are consistent, and the presence of reversals and positive baked contact tests indicate that the magnetization in these rocks was acquired soon after its formation.

Analyzing a great number of paleomagnetic poles from western Laurentia (Slave–Rae–Hearne craton), Irving et al. (2004) separated their poles into two groups: the post-Hudsonian paleopoles with ages between 1740 and 1760 Ma, and the syn-Hudsonian paleopoles with ages around 1835 Ma, although some of these poles have assigned ages as old as 1963 Ma. Most of the post-Hudsonian poles are interpreted as overprinting magnetizations in sedimentary rock units that were subjected to high-grade metamorphism during the Hudsonian orogeny. One peculiar difference between these two groups is that the younger 19 poles are characterized by a single polarity while the older 18 poles are characterized by both polarities. Irving et al. (2004) calculated a mean pole (Table 3) for the younger 1760–1740 Ma poles (20.2°N, 267.7°E, $N = 19$, $K = 42$, $A_{95} = 5.7°$) which includes their well-dated 1740+5/−4 Ma Cleaver dykes pole. This mean pole is plotted in Fig. 16a (L-1750) after rotation to the Nuna configuration. According to Irving et al. (2004), the determination of a 1835 Ma syn-Hudsonian mean pole is more complicated due to tectonic effects (rotations along local vertical axis) and incomplete elimination of secondary components, related to the above mentioned single polarity overprint (1760–1740 Ma), that affect these paleopoles. These authors selected six paleopoles (Table 3) to calculate a mean pole that represents the position of western Laurentia at 1835 Ma age. This pole is located at 1.1°S, 288.3°E, $A_{95} = 12.1°$. This mean pole embodies two tectonically unaffected rock units, the 1827 ± 4 Ma Sparrow dykes and the 1854 ± 11 Ma Wathaman Batholith, whose poles are situated close together, and also the Dubawnt pole mentioned above. Also, a nearly equal number of negative and positive inclinations associated to all six paleopoles helps to eliminate the eventual presence of small amounts of secondary components (Irving et al., 2004). The Irving's et al., 1835 Ma Laurentian pole is also shown in Fig. 16a (L-1835-a) after rotation to the Nuna configuration. Noteworthy is the fact that if we trace an APW path for Laurentia using the 1835 Ma and 1750 Ma means (light grey in Fig. 16a) it passes nearly over the north geographic pole, corroborating the fact that western Laurentia was probably part of the Nuna supercontinent at ca. 1780 Ma ago as in the configuration proposed in Fig. 16a. In this configuration, western Laurentia is close to the equator, which is consistent with hot climate lithologies, such as, carbonates, evaporates, and redbeds as the main constituent of syn-Hudsonian formations (Irving et al., 2004).

In the interpretation of Irving et al. (2004), the Coronation loop firstly considered as a paleomagnetic imprint of the Wopmay orogeny (McGlynn and Irving, 1978) is now attributed to different tectonic rotations along local vertical axis that affected several poles that defined it. However, Mitchell et al. (2010) argue that many poles positions cannot be entirely explained by rotations along local vertical axis. Using the regional tectonic setting of the conjugate Bathurst and the McDonald transcurrent Fault systems (northern Wopmay orogen), they used their equal rotation model (where both fault systems rotated equally and oppositely 12°) to structurally correct poles associated to the Kilohigok and Great Slave basins that pre-date transcurrent faults (Table 3). These corrections bring close together some of the poles, however do not eliminate the east-west displacements of other poles (see Fig. 7c of Mitchell et al., 2010). These authors attribute these displacements (up to 150 cm/year) to rapid, oscillatory true polar wander (TPW) since they cannot be explained by normal continental drifting rates. We used a group of consistent poles (after tectonic correction according to Mitchell et al., 2010, Table 3) dated at 1885–1870 Ma to calculate a mean for western Laurentia (13.7°S; 265.4°E; $N = 10$; $\alpha_{95} = 9.9°$). In principle, this pole (L-1880 in Fig. 16a, after rotation to the Nuna configuration) was not affected by TPW and may represent the position of western Laurentia at that time. Irving et al. (2004) used two of these older poles to calculate their 1835 Ma mean pole for western Laurentia. Excluding these poles we calculated a new 1835 Ma mean pole (4.8°N; 287.0°E; $\alpha_{95} = 13.4°$) for western Laurentia using the following poles: the 1827 ± 4 Ma Sparrow dykes, the 1854 ± 11 Ma Wathaman Batholith, the 1830–1760 Ma Dubawnt Group, and the 1818 ± 4 Martin Formation. This pole (L-1835-b in Fig. 16a) is not far from the Irving's et al., 1835 pole (L-1835-a in Fig. 16a, after rotation to Nuna configuration), and we can trace an APW path between 1880 Ma and 1750 Ma for Laurentia (dark grey in Fig. 16a). This new APW path also passes nearly close the geographic pole at the time between 1835 and 1750 Ma ago for Laurentia in the Nuna configuration of Fig. 16a.

Table 3
Selected paleomagnetic poles from Laurentia, Baltica, North China and Amazonian Craton.

Continent/formation	Plat (°N)	Plong (°E)	A_{95} (°) *dp*/*dm*	Euler pole[e]	#RPlat (°N)	#RLong (°E)	RPlat (°N)	RLong (°E)	Age (Ma)	Ref.
Laurentia				0.3°N, 7.2°E (−83.4°)						
1 – Kilohigok sill	−27	268	–		−25	282	58	285	1870 ± 4	1
2 – Peacock Hills	−15	270	11		−13	283	70	293	1882 ± 4	1
3 – Mara	−07	253	07		−06	266	73	235	1885 ± 5	1
4 – Peninsula sill	−22	263	07		−22	263	58	251	1870 ± 4	1
5 – Takiyuak	−13	249	08		−13	249	56	222	1885–1870	1
6 – Douglas Peninsular	−17	258	16		−17	245	50	224	1885–1870	1
7 – Akaitcho River	−04	268	07		−05	256	66	215	1885 ± 5	1
8 – Seton (C)	02	267	06		01	256	68	200	1885 ± 5	1
9 – Pearson (A)	−19	283	09		−22	269	60	261	1870 ± 4	1
10 – Kahochella (K)	−07	298	09		−11	286	71	303	1882 ± 4	1
Mean (L-1880) [a]	−14	265	10						1880	2
11 – Sparrow dykes	12	291	08				76	030	1827 ± 4	3, 4
12 – Wathaman Batholith	09	293	04				74	017	1854 ± 11	5
13 – Dubawnt group (L-1780)	07	277	08				90	144	1785 ± 4	6
14 – Martin	−09	287	09				72	308	1818 ± 4	1
Mean (L-1835a)[b]	−01	288	12				77	332	1835	7
Mean (L-1835b)[c]	05	287	13				80	357	1835	2
Mean (L-1750)[d]	20	268	06				73	131	1750	7
15 – Spokane formation	−16	226	05						1460	8
16 – Michikamau intrusives.	−02	218	3/7						1460 ± 5	9
17 – Harp Lake Complex	02	206	3/6						1450 ± 5	10
18 – St. Francois Mountains	−13	219	07						1476 ± 16	11
19 – Tobacco Root Mountains	09	216	10						1448 ± 49	12
Mean (L-1460)	−04	217	12				29	195	1460	
20 – Laramie anorthosite (L-1429)	−07	215	04				26	198	1429 ± 9	13
21 – Electra Lake Gabbro (L-1433)	−21	221	03				28	215	1433 ± 2	14
22 – MacKenzie dykes (L-1265)	04	190	05				03	183	1267 ± 7/−3	15, 16
Baltica				27.8°N, 163.2°E (56.3°)						
23 – Shosksha Form. (B-1780-S)	42	221	07				77	166	1790–1770	17
24 – Ropruchey sills (B-1770-R)	41	230	08				83	176	1770 ± 12	18
25 – Tuna dykes	21	180	07						1461–1462	19
26 – Salmi Form. vulc. rocks	06	200	11						1460	19
27 – Lake Ladoga dolerites	16	177	04						1457 ± 2	19
28 – Valaam monzodioritic – A pole	14	166	02						1458 ± 4	19
29 – Lake Ladoga mafic rocks	15	177	6						1452 ± 12	20
Mean (B-1460)	15	180	13				33	186	1460	
29 – Vaasa dykes	07	164	04						1228 ± 13	19
30 – Satakunta dykes and sills	02	158	04						1264 ± 12	19
31 – Märket dolerites	−06	146	09						1265 ± 6	19
Mean (B-1265)	04	158	04				09	180	1265	
North China Craton				39.8°N, 310.5°E (−68.1°)						
32 – Taihang dykes (NC-1770)	36.0	247.0	03				86	237	1769 ± 3	21
33 – Yangzhuang dol. (NC-1350)	2.4	190.4	12				29	130	1350	22
34 – Yunmengsham Fm. (NC-1270)	60.6	267.0	04				64	351	1270	
Amazonian Craton				77.4°N, 93.0°E (128.9°)						
35 – Colider Suite (AC-1790)	63	119	10				86	238	1789 ± 7	23
36 – Nova Guarita dykes (AC-1419)	−48	246	07				−57	346	1419 ± 4	2

Notes: Plat – pole latitude; Plong – pole longitude; A_{95}, *dp*, *dm* – Fisher's statistic parameter. Refs. 1 – Mitchell et al. (2010); 2 – this work; 3 – McGlynn et al. (1974); 4 – Bostock and van Breemen (1992); 5 – Symons (1991); 6 – Park et al. (1973); 7 – Irving et al. (2004); 8 – Vitorello and Van der Voo (1977); 9 – Emslie et al. (1976); 10 – Irving et al. (1977); 11 – Meert and Stuckey (2002); 12 – Harlan et al. (2008); 13 – Harlan et al. (1994); 14 – Harlan and Geissman (1998); 15 – Buchan and Halls (1990); 16 – LeCheminant and Heaman (1989); 17 – Pisarevsky and Sokolov (2001); 18 – Fedotova et al. (1999); 19 – Salminen and Pesonen (2007); 20 – Halls et al. (2000); 21 – Pei et al. (2006); 22 – Bispo-Santos et al. (2008). #Plat/#Plong – paleolatitude/paleolongitude of Laurentian poles rotated according to Mitchell et al. (2010) (see text for details). RPlat/RPlong – rotated pole latitude/pole longitude.

[a] This mean was calculated using poles from 1 to 10.

[b] This mean was calculated using poles from 9 to 14.

[c] This mean was calculated using poles from 11 to 14.

[d] This mean was calculated using 19 post-Hudsonian poles after Irving et al. (2004).

[e] Euler rotation poles used to rotate poles and continents to the Nuna configuration (Fig. 16a).

Based on paleo-trade wind directions during the syn-Hudsonian sedimentation (Hildebrand, 1988; Hoffman and Grotzinger, 1993), Irving et al. (2004) concluded that Laurentia was in the northern hemisphere at that time. This fact can be used to constrain the position of the other continental blocks as shown in Fig. 16a.

9.2. Longevity of the Nuna supercontinent

A paleomagnetic test of the longevity of the Nuna supercontinent paleogeography proposed by Bispo-Santos et al. (2008) is limited to very few periods where key paleomagentic poles are available for two or more continental blocks that formed that supercontinent. Recently, Salminen and Pesonen (2007) and Lubnina et al. (2010) have shown that Laurentia and Baltica maintained their relative position up to ~1270 Ma ago, based on comparisons of key poles from both continents at 1460 Ma and at 1270 Ma, respectively. Fig. 16b shows mean poles (Table 3) calculated for these two cratons for ca. 1460 Ma and ca. 1265 Ma, respectively, after rotating them according to the Nuna configuration of Fig. 16a. These mean poles are essentially the same as described by Salminen and Pesonen (2007). However, for the 1460 Ma Laurentian mean pole we included the recently published pole (TR) obtained for the Tabacco Root Mountains (Harlan et al., 2008). Its respective dual-polarity A_1 component is considered to be primary by the authors although two baked contact tests yielded inconclusive results since no different stable directions were found for the country rocks (gneisses) collected far from the dyke. An age of 1448 ± 49 Ma was obtained for one of the studied dykes (Sm–Nd methodology), which is considered the best estimated age for pole TR. Also, this pole is similar to other ca. 1450 Ma poles from Laurentia. For the 1460 Ma Baltica mean pole, we included the recent published pole (LL) obtained for the Lake Ladoga region (Lubnina et al., 2010). U–Pb dating on baddeleyite yielded a crystallization age of 1452 ± 12 Ma, and positive baked contact and reversal tests attest the primary character of the magnetization. Note that the respective mean poles for Laurentia and Baltica at 1460 Ma and 1270 Ma in Fig. 16b overlap within error.

Until now, the absence of key paleomagnetic poles of similar ages from North China and Amazonia did not permit a test of the longevity of the Nuna Supercontinent relative to these two cratons (Fig. 16a). The well-dated 1419 Ma Nova Guarita pole presented here gives the basis for such a test. This pole is plotted in Fig. 16b (AC-1419), after rotating it according to the Nuna configuration. Since this pole is somewhat younger than the 1460 Ma Baltica and Laurentian poles, we also plotted in this figure two other Laurentian poles of similar ages (after rotation to the Nuna configuration). These poles are, respectively, the Laramide anorthosite/Sherman granite (LA) pole with an age of 1429 ± 9 Ma (Harlan et al., 1994) and the Electra Lake Gabbro (ELG) pole with an age of 1433 ± 2 Ma (Harlan and Geissman, 1998). The Nova Guarita pole plots somewhat different from the Baltica and Laurentia poles, which may suggest that the Amazonian Craton started to break up from Nuna at that time.

Unfortunately, no reliable paleomagnetic pole is presently available for North China for this period. The best results from this craton were obtained for Mesoproterozoic sediments (dolostones) from the Jixian region (North China) (Pei et al., 2006). ChRM directions yielded a positive fold test, which suggests they represent primary magnetizations. An age ca. 1350 Ma was assigned to the corresponding paleomagnetic pole (NC-1350 in Fig. 16b after rotation to Nuna configuration), which is ~70 Ma younger than the Nova Guarita pole. For younger periods, a paleomagnetic pole was obtained for the Yunmengshan Formation which has Rb–Sr age of ca. 1270 Ma (Zhang et al., 2006). The analyzed rocks (purple mudstones, muddy sandstones and andesites) yielded a high-unblocking temperature component carried by hematite (sediments) and magnetite (andesites). This component passes both the reversal and the fold tests, and is interpreted to be primary (Zhang et al., 2006). Fig. 16b shows the Yunmengshan pole (NC-1270) after rotation to Nuna configuration. Note that this pole is very different from similar-age poles of Laurentia and Baltica, suggesting that North China had also drifted away from the Nuna supercontinent at that time.

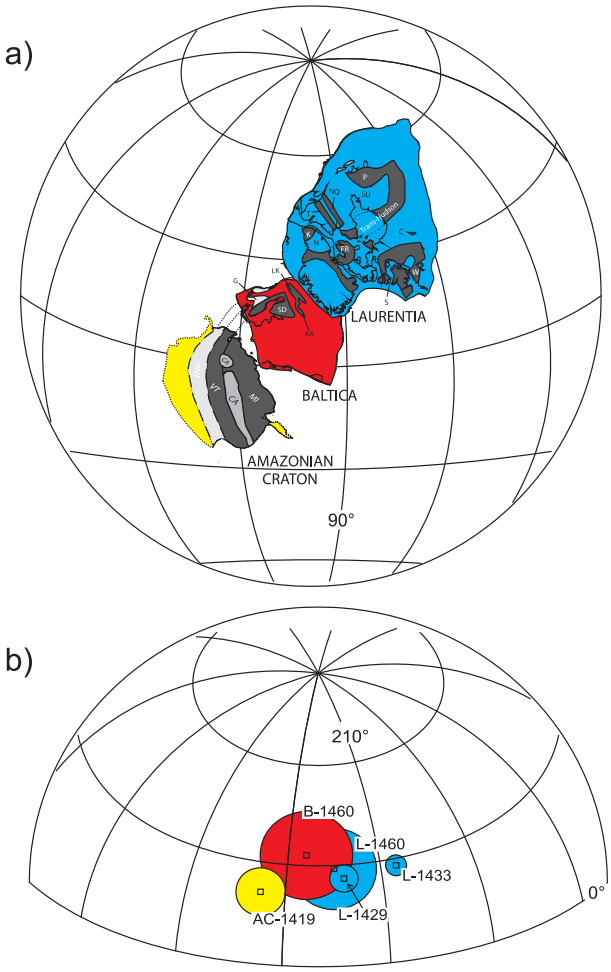


Fig. 17. (a) Nuna reconstruction with Amazonia and Baltica as in the SAMBA model (Johansson, 2009). Laurentia and Baltica were rotated as in Fig. 16a. Amazonia was rotated using the Euler rotation pole: 70.3°N, 167.5°E, 126°. (b) Paleomagnetic poles (Table 3) were rotated using the same Euler rotation poles assigned for each continent. Geological provinces as in Fig. 16a. See text for details.

Other paleogeographic configurations propose however a connection between Baltica and Amazonia at the Mesoproterozoic. In several geological models of Nuna, the north-northwestern (present) Amazonian Craton is linked directly to southwestern (present) Baltica (Geraldes et al., 2001; Zhao et al., 2002, 2004, 2006; Hou et al., 2008a,b; Johansson, 2009). The 1.79 Ga Colider pole of Bispo-Santos et al. (2008) precludes such a configuration for the Paleoproterozoic, but the 1.42 Ga Nova Guarita pole allows us to test the Laurentia–Baltica–Amazonia link at Mesoproterozoic times. Fig. 17a shows Amazonian Craton linked to Baltica in a similar configuration to that proposed in the SAMBA model (Johansson, 2009). Laurentia and Baltica were rotated as in Fig. 16a, and we have used the following Euler rotation pole for Amazonia: 70.3°N, 167.5°E, 126°. Fig. 17b shows the 1.46–1.42 Ga poles from these three cratonic blocks after rotating them using the same Euler rotation poles assigned for each continent. Note that the confidence circle of the Nova Guarita pole overlaps the confidence circle of the 1460 Ma mean pole of Baltica. Given the uncertainties in the

ages and position of all these poles, we can say that they favor the configuration of Nuna as shown in Fig. 17 a at Mesoproterozoic times. One possibility is that the Colider Suite pole does not represent the position of the proto-Amazonian Craton at 1.78 Ga ago. So, only with new Paleo- to Mesoproterozoic poles from the Amazonian Craton we can safely establish the position of the Amazonian Craton in the Nuna configuration.

10. Conclusions

Paleomagnetic analysis associated with ^{40}Ar–^{39}Ar geochronology of the Nova Guarita Intrusive dykes (northern Mato Grosso State, Brazil) resulted in a well-dated 1418.5 ± 3.5 Ma paleomagnetic pole for Amazonia (NG pole), located at 245.9°E, 47.9°S (A_{95} = 7.0°). Magnetic mineralogy, petrographic analyses and positive baked-contact test suggest that the NG pole represents a primary thermo-remanent magnetization, which can be classified with a reliability factor Q = 6. Comparison of this pole with the present Mesoproterozoic paleomagnetic database for Laurentia, Baltica and North China suggests that Amazonian Craton might have started to break up from the Paleoproterozoic Nuna Supercontinent at around 1420 Ma consistent with the proposal of Bispo-Santos et al. (2008). Other paleogeographies however may accommodate the new paleomagnetic results obtained here into a Mesoproterozoic Nuna supercontinent.

Acknowledgements

We thank: (i) Companhia Matogrossense de Mineração (METAMAT) for its help in the logistic field work, (ii) the financial support from FAPESP (Grants 07/53177-4, 07/59531-4 and 11/50887-6) and CNPq (Grant 554458/2005-5), (iii) GEOCIAM (Instituto Nacional de Ciência e Tecnologia de Geociências da Amazônia) and CAPES for the financial support of the co-author Liliane Janikian, (iv) Maria Anna Abreu de Almeida dos Reis by the hysteresis parameters determination and the University of Southampton – National Oceonography Centre (NOC-England) where these measurements were done.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.precamres.2011.10.022.

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